Inference for Moment Inequalities with Nuisance Functions

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This paper proposes a conceptually novel semiparametric model formulated by moment inequalities, which involve partially identified finite-dimensional parameters of interest and point identified infinite-dimensional nuisance parameters. We propose a reparametrized GMM condition to obtain the variance estimator by a corresponding influence function. We partially identify the parameters of interest through their confidence set. We establish the uniform bootstrapping consistency and the uniform asymptotic size control for the constructed confidence set. We illustrate the method by a structural entry game model with partially identified state-dependent interaction effects, where the true conditional choice probabilities serve as the nuisance functions.

KEYWORDS: partial identification, semiparametrics, moment inequalities, uniform asymptotic size, bootstrap, conditional choice probability, structure model

1 Introduction

This paper addresses with a novel semiparametric model formulated by a finite number of unconditional moment inequalities with partially identified finite-dimensional parameters of interest and point identified infinite-dimensional nuisance parameters, which has not been generally resolved in the

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literature. This paper adopts a non-standard two-step testing method to conduct inference on partially identified parameters of interest by the confidence set of the true values of parameters of interest.

Under an unconditional moment inequality model, when the parameters of interest are partially identified without any nuisance parameters, the usual sample mean and sample variance of the moment functions, as the estimator of the asymptotic variance of the sample mean, can be applied. However, when the nuisance parameters are present and are approximated by their estimators, the variance estimator might be different from the usual sample variance. The confidence set of interest regarding the test concerning the moment inequalities will then be affected by the inconsistent variance estimator. And the size of the test cannot be controlled. Therefore, the fundamental question we need to deal with is the variance estimator in a moment inequality model where there are nuisance parameters, particularly infinite-dimensional nuisance parameters.

In this paper, we first assume the point identification of the nuisance parameters taking the form of the mean square projections and find its estimators. Through a reparametrized GMM condition, we derive the general formula of the sample mean of the moment functions and the estimator of the asymptotic variance of the sample mean by the corresponding influence function, applying the results from Newey (1994a). The reparameterized GMM condition helps to transform the complex moment inequalities to moment equalities so that the pathwise derivative can play a role. These estimators given the estimators of the nuisance parameters appreciate the effect of the nuisance parameters on the confidence set of interest.

The confidence set of interest is then formed by a two-step procedure following Romano et al. (2014). The first step is to form the confidence set of population mean of moment functions μ_M, in order to deal with the slackness parameter $\sqrt{n}\mu_M$ that cannot be consistently estimated. The second step is to construct the confidence set of interest given μ_M, which takes the value of the lower-left vertex of its confidence set, along with a Bonferroni-type correction. Under this two-step procedure, the probability that the estimator of the nuisance parameters violates the moment inequalities when the moment inequalities hold is considered through the first step confidence set, along with the aforementioned estimators. The test corresponding to the Bonferroni confidence set is then shown to sat-

isfy the uniform asymptotic size control (UASC) property. The result relies on the uniform bootstrapping consistency when the nuisance parameters are present in the test statistics.

The method has general implications in all fields of economics. For example, a semiparametric GMM model with partially identified average treatment effect and point identified propensity score. In this paper, we illustrate the method by an interval outcome regression model based on a partial linear model as dicussed in the simulation study.

We also demonstrate through the Conditional Choice Probability (CCP), that are point identified nuisance functions occurring in moment inequality models of industrial organization and microeconometrics, which has not been fully addressed with in the literature. We develop a novel complete structure model of a static discrete incomplete information game with partially identified state-dependent interaction effects among radio stations. The construction of the structure model for the payoff-related parameters of interest involved in strategic behaviors of radio stations consists of three stages. The first stage is to find the unique Bayesian Nash Equilibrium (BNE) upon the information structure. The second stage is to adopt the BNE to form a unique prediction model in terms of CCP. The third stage is to establish the econometric model in terms of moment inequalities via taking the expectation of the prediction model written in terms of inequalities. The confidence set of interest is then founded by grid search.

The main contribution of this paper is the development of a novel semiparametric framework in terms of the unconditional moment inequality model, that allows point identified infinite dimensional nuisance parameters along with partially identified finite dimensional parameters of interest. It enriches the literature of semiparametrics by dealing with partially identified parameters of interest and that of partial identification by dealing with infinite-dimensional nuisance parameters. The paper provides the reparameterized GMM condition for the general unconditional moment inequality model with nuisance functions and the consequential derivation of the asymptotic variance of the sample mean of the moment functions. Besides, the paper contributes to statistics by the derivation of the uniform bootstrapping consistency in the presence of nuisance functions. And it also contributes to the applied research by the structure model or

the general solution to deal with the CCP as the nuisance function occurring in the moment inequality models. The main caveat however is we do not provide the identification condition of the nuisance parameters while we assume it to be point identified, which is not the focus of this paper. Besides, the further reduction of computation burdens will be an interesting topic in the future.

When we are facing with moment equality conditions and focusing on the (possibly over-) identified parameters of the parametric part but with some identified nonparametric nuisance parameters, the semiparametric model is the general method to infer the parameters of interest in the parametric part. The rich literature includes Robinson (1987), Robinson (1988), Andrews (1994), Hahn (1998), Hirano et al. (2003), Ackerberg et al. (2014) and Chernozhukov et al. (2022), etc. However, this paper generalizes the semiparametric model into the form of moment inequalities with partially identified parameters of interest.

When it comes to moment inequality conditions while the parameters of interest are usually not point identified, even without the nuisance parameters, the usual M-estimation might not be applicable unless strong assumptions imposed[1]. Meanwhile, using any elements of the identified set to estimate the parameters of interest is impossible to achieve consistency, since the value in the identified set cannot be distinguished from the true value of parameters of interest. And the general estimator of the identified set is hard to guarantee the set consistency, such as Hausdorff consistency. The usual way to deal with moment inequalities to conduct inference on partially identified parameters of interest thereby turns to focus on the confidence set, under some hypothesis testing procedures.

One approach to the confidence set is the confidence set of the identified set of parameters of interest discussed by Manski and Tamer (2002), Chernozhukov et al. (2007), Beresteanu and Molinari (2008), Romano and Shaikh (2010), Beresteanu et al. (2011) and Luo et al. (2024). Another approach focuses on the confidence set of the true parameter value promoted by Imbens and Manski (2004), Andrews and Soares (2010a), Andrews and Barwick (2012) and Romano et al. (2014). In addition to unconditional moment inequality models, the conditional moment inequality models have been developed relatively afterwards by

[1] Tamer (2003) discusses the point identification of parameters of interest in a moment inequality model.

Andrews and Shi (2013), Chernozhukov et al. (2013), Andrews et al. (2023) and Cox and Shi (2023).

In this paper, we focus on the confidence set of the true parameter value that applies to more general identified sets which are not necessarily convex and that could be typically constructed smaller. In particular, we follow Romano et al. (2014) to achieve the UASC property for testing the moment inequalities with nuisance functions. However, Romano et al. (2014) does not consider the model with nuisance parameters.

When there are nuisance parameters occurred in moment inequalities while the parameters of interest are partially identified, Shi and Shum (2015) develops a two-stage estimation method to deal with the point-identified nuisance parameters. For partially identified nuisance parameters, Kaido et al. (2019) considers an unconditional moment inequalities and partially identifies the subvector of parameters by the projection of confidence sets. Cox and Shi (2023) and Andrews et al. (2023) consider the conditional moment inequality models with nuisance parameters entering linearly and partially identify the parameters of interest by different methods of adaptive testing, that drastically reduce the computation burden. However, all these literatures for partially identified subvector parameters in the moment inequalities deal with finite-dimensional nuisance parameters only.

In this paper, we consider a general form of moment inequalities with nuisance parameters that are infinite-dimensional. We assume some regularity conditions, such as Lipschitz continuity, regarding the nuisance parameters. Besides, we assume the data generating process (DGP) satisfiies the uniform integrability condition. The confidence set of parameters of interest could be derived by inverting the test, in particular, bootstrapping the test statistics in two steps and then following a grid search procedure or some other proper algorithms.

The remainder of the paper is organized as follows. Section 2 discusses the moment inequality model and shows the uniform asymptotic size control. Section 3 provides the algorithm manual and the simulation study of an interval outcome regression model. Section 4 discusses an empirical application under a general game-theoretical structure model for radio stations in the U.S. Section 5 concludes.

2 Moment Inequalities

In this paper, we consider the true value θ_0 and h_0 that satisfy the moment inequalities:

$$E_{F_0}[m_j(W_i, \theta_0, h_0)] \geq 0, j = 1, \ldots, k \tag{1}$$

where $\theta_0 (\in \Theta \subset \mathbb{R}^d)$ is the finite-dimensional parameter of interest and $h_0 (\in \mathcal{H} \subset \mathcal{L}^2(x))$ is the infinite-dimensional nuisance parameter. $X = x$ is a subvector of W that will be defined accordingly under a specific moment inequality model. $\{m_j(\cdot, \theta, h) : j = 1, \ldots k\}$ are known real-valued moment functions. The observed random sample of random vectors $W_i (i = 1, \ldots n)$ are assumed to be i.i.d. throughout the paper with the true joint distribution $F_0 \in \mathbf{F}$ on $\mathbb{R}^p$, where $\mathbf{F}$ is a class of nonparametric distributions. We consider W_i defined based on Lebesgue measure. The moment inequalities (1) are general enough since it could be adopted to form moment conditions for moment functions that less or equal to zero and thereby for moment equalities. Define the generics $(\theta, h, F) \in \mathcal{F}$ that satisfies (1), (26) and the RC assumptions in the appendix for $\theta_0 = \theta$ and $h_0 = h$. The key feature of this model is that θ_0 need not be point-identified, whereas h_0 is assumed to be point-identified under the form of mean square projections and the identification of h_0 is not the focus of this paper. That is, given the true DGP F_0, we could pin down $h_{0,}$ however we could not pin down θ_0 unless additional sufficient information is provided.

In this model, there are two main tasks in inferring θ_0. The first one comes from the nuisance parameter h_0 that will affect the inference of θ_0, so that we need to estimate it first. In this paper, we are interest in point identified h_0 and estimating it by mean square projection, i.e. $h_0(x) = E(Y|X)$, where both X and Y are subvectors of W that will be determined under the specific models. We then characterize the impact of first stage estimation of the nuisance parameter h_0 on the second stage inference of the parameter of interest θ_0 by the pathwise derivative under an reparametrized GMM condition of (1) following Newey (1994a).

The second one comes from the inequality of the moment conditions (1), which implies that θ_0 is not necessarily point identified. In this paper, we are interested in partially identified θ_0 by getting a confidence set covering the true parameter θ_0 in the end, rather than forming additional conditions to point iden-

tify θ_0. We will achieve it in a two-step procedure following Romano et al. (2014) where the first step is to get the confidence set of population mean of the moment functions or μ_M in (1) and the second step is to get the confidence set of θ_0. However, the critical values are data-dependent and thereby depend on the true DGP, therefore the uniform asymptotic size control (UASC) regarding the confidence set of θ_0 is crucial and thereby demanded.

In the first stage, the consistent and asymptotic normal (CAN) estimator of h_0, i.e. $\widehat{h}_0 = \widehat{E(Y|X)}$, could be obtained through the kernel estimation or series estimation such that $\gamma_n(\widehat{h}_0 - h_0) \xrightarrow{d} \mathcal{N}(0, V_h)$. As the Assumption 3 in the appendix, it is sufficient to require that the convergence rate of $\widehat{h}_0$ is faster than $n^{-1/4}$ in terms of norm $\| h \|$ in this paper. Through the inference based on the sub-model of the DGP F_τ in the direction of τ, we could get the pathwise derivative of the population mean of the moment functions or μ_M in (1) denoted as a $k \times 1$ vector d(W) which measures the influence of $\widehat{h}_0$ on the second stage confidence set of θ_0 through the estimator of μ_M.

Given $\widehat{h}_0$, we want to conduct inference on partially identified θ_0 such that the system (1) is satisfied. In order to avoid the potential Hausdorff inconsistency issue of the identified set (IDS), that is $\Theta_0 = \{\theta \in \Theta : (1)\}$, and to fit more general IDS which might not be convex, we focus on the inference of θ_0 using the confidence set of θ_0:

$$CS_n = \{\theta \in \Theta : T_n(\theta, \widehat{h}_0)) \leq C_n(1-\alpha, \theta, \widehat{h}_0)\}, \tag{2}$$

while we consider testing

$$H_0: \theta_0 = \theta \; s.t. \; E_{F_0}[m_j(W_i, \theta_0, h_0)] \geq 0, j = 1 \ldots k, \tag{3}$$

where the alternative is $H_1 : \theta_0 = \theta \; s.t. \; E_{F_0}[m_j(W_i, \theta_0, h_0)] < 0$ for at least one $j \in \{1, \ldots, k\}$ and $C_n(1-\alpha, \theta, \widehat{h}_0)$ is the $1-\alpha$ quantile of the asymptotic distribution of $T_n(\theta, \widehat{h}_0) \equiv T_n(\theta)$, where $\alpha \in (0,1)$. The idea of the confidence set derives from the duality of hypothesis testing and confidence set.

As pointed out by Andrews and Soares (2010a), there is a general class of statistics in the form of $T_n = S(\sqrt{n}\widehat{\mu_n}(\theta), \widehat{\Sigma_n}(\theta))$ which includes

$$T_n^{max} = \underset{1 \leq j \leq k}{max} - \frac{\sqrt{n}\widehat{\mu_{j,n}}}{S_{j,n}}, \tag{4}$$

$$T_n^{qlr} = \inf_{t \in \mathbb{R}^k_{+,\infty}} Z_n(t)' \widehat{\Omega}_n^{-1} Z_n(t), \tag{5}$$

where $Z_n(t) = \big(\frac{\sqrt{n}(\widehat{\mu_{1,n}} - t)}{S_{1,n}}, \ldots, \frac{\sqrt{n}(\widehat{\mu_{k,n}} - t)}{S_{k,n}}\big)$,

$$T_n^{qlr,ad} = \inf_{t \in \mathbb{R}^k_{+,\infty}} Z_n(t)' \widetilde{\Omega_n}^{-1} Z_n(t), \tag{6}$$

where $\widetilde{\Omega_n} = \widehat{\Omega_n} + max\{\epsilon - det(\widehat{\Omega_n}), 0\} \cdot I_k$, and

$$T_n^{mmm} = \Sigma_{j=1}^k \big(\frac{\sqrt{n}\widehat{\mu_{j,n}}}{S_{j,n}}\big)^2 \cdot \mathbb{1}\{\widehat{\mu_{j,n}} < 0\}. \tag{7}$$

The function $S : \mathbb{R}^k \times \mathbb{R}^{k \times k} \to \mathbb{R}$ is continuous in both arguments and weakly decreasing in each element of its first arguments. Besides, $S(m, \Sigma) = S(Dm, D\Sigma D)$ for all $m \in \mathbb{R}^k$, $\Sigma \in \mathbb{R}^{k \times k}$ and positive definite $D \in \mathbb{R}^{k \times k}$. In this paper, let $\mu_{M_j} \equiv \mu_{M_j}(\theta, h_0, F) \equiv E_F[m_j(W_i, \theta, h_0(x))]$, $j = 1, \ldots, k$, $\Sigma \equiv \Sigma_M(\theta, h_0, F) \equiv V_F[m(W_i, \theta, h_0(x))]$, $D \equiv Diag(\Sigma) = D(\theta, h_0, F)$, $\Omega \equiv \Omega(\theta, h_0, F) \equiv D^{-\frac{1}{2}} \Sigma D^{-\frac{1}{2}}$, $\sigma_j^2(\theta, h_0, F)$ denotes the j-th diagonal element of Σ, $\widehat{\mu}$ denotes the estimator of μ_M, $\widehat{\Sigma}$ denotes the estimator of Σ and similarly for $\widehat{D}$ and $\widehat{\Omega}$ and $S_{j,n}^2 \equiv \sigma_j^2(\theta, \widehat{h_0}, \widehat{F_n})$ where $\widehat{F_n}$ denotes the empirical distribution of the $W_i (i = 1, \ldots, n)$, that is $\widehat{F_n}(w) = \frac{1}{n}\Sigma_{i=1}^n \mathbb{1}\{\{W_i \leq w\}$, where $\mathbb{1}\{\cdot\}$ denotes the indicator function. Hence, $\widehat{\mu_j} = \widehat{\mu_{j,n}} = \mu_{M_j}(\theta, \widehat{h_0}, \widehat{F_n})$ and $\widehat{\Sigma} = \Sigma_M(\theta, \widehat{h_0}, \widehat{F_n})$.

$$\begin{aligned} T_n(\theta) &= S(\sqrt{n}\widehat{\mu}(\theta), \widehat{\Sigma}(\theta)) \\ &= S\big(\sqrt{n}\widehat{D}^{-\frac{1}{2}}(\theta)(\widehat{\mu}(\theta) - \mu_M(\theta)) + \sqrt{n}\widehat{D}^{-\frac{1}{2}}(\theta)\mu_M(\theta), \widehat{\Omega}(\theta)\big) \end{aligned} \tag{8}$$

When we try to find the asymptotic distribution of $T_n(\theta)$, under suitable conditions, the inner part $\sqrt{n}\widehat{D}^{-\frac{1}{2}}(\theta)(\widehat{\mu}(\theta) - \mu_M(\theta))$ is convergent in distribution to a normal distribution while $\widehat{\Omega}(\theta)$ is convergent in probability to a correlation matrix. However, since $\sqrt{n}\mu_M$ cannot be consistently estimated, we may want to replace it by some statistics such that our ultimate goal that the test defined by the critical function ϕ_n later using (27) is uniformly valid and the UASC regarding the confidence set of θ_0 w.r.t. the approximated critical value $\widehat{C_n(1-\alpha, \theta, \widehat{h_0})}$ of

the test is achieved, that is,

$$\limsup_{n\to\infty} \sup_{(\theta,h,F)\in\mathcal{F}} E_F[\phi_n] \leq \alpha. \tag{9}$$

We will obtain the critical value by bootstrapping the test statistics after the replacement demonstrated by a two-step procedure. As pointing out by Andrews (2000), we cannot simply bootstrap $T_n(\theta)$ to get its critical value and to form the test. This is because the bootstrapping may fail when the true DGP is on the boundary of the DGP space while we need a uniform consistency in levels that is defined in this paper as (9). The UASC is crucial because $\widehat{C_n(1-\alpha,\theta,\widehat{h}_0)}$ is data-dependent and thereby depend on the true DGP so that the objective size need to be correct for the sample size regardless of the DGP in order to have meaningful asymptotic properties.[2] For more details regarding the importance of the uniformity, Imbens and Manski (2004) discussed an example of finite sample sample size distortion and Andrews and Guggenberger (2009) discussed the possible discontinuity of the asymptotic distribution of $T_n(\theta)$ under drifting sequence of parameters.

In this paper, we follow Romano et al. (2014) to adopt a two-step procedure to conduct uniform inference on the partially identified θ_0 through bootstrapping. Andrews and Soares (2010a) considers a method where $\sqrt{n}D^{-\frac{1}{2}}\mu_M$ is replaced by a moment selection function $\varphi(\xi,\Omega)$ defined there to avoid the consistent estimation for $\sqrt{n}\mu_M$ and to find the critical value $\widehat{C_n(1-\alpha,\theta)}$ so that the UASC property is achieved. Similar to Andrews and Soares (2010a), the key replacement applying the method in Romano et al. (2014) for $\sqrt{n}\mu_M$ is thereby the lower-left vertex of the confidence set of μ_M multiplied by root n, denoted as $\sqrt{n}\lambda \in \mathbb{R}^k$. This confidence set about $\sqrt{n}\mu_M$ will be the first step confidence set corresponding to the test $H_{0'}$ defined by (12). Let $\widehat{C_n(1-\alpha,\theta,\widehat{h}_0)}$ be the $1-\alpha$ quantile of the asymptotic distribution of $T_n^{TS}(\theta)$ which denotes the $T_n(\theta)$ after the replacement. The second step confidence set corresponding to the test H_0 will then be that of θ_0 given μ_M defined by (23). The UASC corresponding to the Bonferroni confidence set will

[2] In another words, the limit of the pointwise size will be very misleading since it does not rule out the case, for example, that for any sample size n, in particular for any $n < inf\{N_1,\ldots,N_I\}$ where I is the cardinality of the DGP space and N_j is the infimum of the set of the natural numbers of sample size of the test w.r.t. the j-th DGP that makes the size controlled, there is a F such that $E_F[\phi_n] \gg \alpha$. The counter-example can be formed using a t-test.

then be shown under these two step testings. This non-standard test procedure is also applied in Staiger and Stock (1997) under a weak IV context. Both $T_n(\theta)$ and $T_n^{TS}(\theta)$ depend on $\widehat{h}_0$ but ignored for notational conciseness.

Therefore, the first step is to construct a confidence set of μ_M under $H_{0'}$ and then a confidence set of θ_0 under H_0 in the second step. Without loss of generality, we will use

$$T_n^{min} = \min_{1\leq j\leq k} \frac{\sqrt{n}(\mu_{M_j}(F) - \widehat{\mu}_{j,n})}{S_{j,n}} \tag{10}$$

as the first step test-statistic and we could use any type of $T_n(\theta)$ as described above as the second step one. Those statistics will typically have asymptotic distributions that are continuous and strictly increasing, which is easy to establish consistency upon getting the critical values through bootstrapping. The uniform bootstrapping consistency of T_n^{min} will be shown upon the Lemma 7, which essentially applies the Theorem 2.4 of Romano and Shaikh (2012a).

Suppose μ_M is the true value $E_{F_0}[m_j(W_i, \theta_0, h_0)]$ and

$$H_{0'} : \mu_M = \mu \, s.t. \, E_{F_0}[m_j(W_i, \theta_0, h_0)] \geq 0, j = 1 \ldots k \tag{11}$$

where $H'_{0'} : \mu_M = \mu \, s.t. \, E_{F_0}[m_j(W_i, \theta_0, h_0)] < 0$ for at least one $j \in \{1, \ldots, k\}$ which enables us to form the test of H_0 conditional on μ_M characterized by (3). The ultimate test of interest defined by (27) will then be constructed along with the test of H_0 and that of $H_{0'}$. The test cannot be rejected when there exists a null of H_0 and $H_{0'}$ is true. The type I error will then be the probability of both H_0 and $H_{0'}$ are rejected simultaneously when there is at least one null is true. Hence, similar to Bonferroni correction for controlling the probability of one or more false rejections not exceed a given level $\alpha \in (0, 1)$ by $\beta \in (0, \alpha)$, we need to introduce a Bonferroni-type correction β here to account for the possible case that μ_M does not lie in $M_n(\beta)$ defined by (12), which may lead to the non-convergence of the first step confidence set as $\mu_M \to \infty$, and to adjust it in the second step. And the test ϕ_n needs to be defined accordingly. In this case, the corrected level β for $H_{0'}$ need to be strictly less than α. Given the Bonferroni-type correction β in the test of $H_{0'}$ and H_0, the probability that the moment inequalities are violated while approximating the nuisance parameters by $\widehat{h}_0$ when the moment inequalities actually hold will be naturally contained in the size of the test of interest. As long as the UASC property holds, this special approximation error will be controlled.

For more generous construction of Bonferroni critical value, see McCloskey (2017). Similar to Andrews and Barwick (2012), β is essentially a tuning parameter under this context and could be found accordingly by maximizing the weighted average power, albeit not the focus of this paper. The naive choice of β is $\frac{\alpha}{10}$ that leads to good power properties as pointed out by Romano et al. (2014). Ideally, the size would be less conservative and the power would be better if β is closer to zero which could be demonstrated by the simulation study in the next section.

The idea of using the lowest λ concerns the least favorable case such that the power is least, that is when $T_n^{TS}(\theta)$ is highest. The first step confidence set is thereby defined by

$$M_n(\beta) = \left\{\mu \in \mathbb{R}^k : T_n^{min} \geq K_n^{-1}(\beta, \hat{F}_n)\right\}, \tag{12}$$

where $\beta \in (0, \alpha)$ and the distribution of T_n^{min} is defined by

$$K_n(x, F) = P\left\{\min_{1\leq j\leq k} \frac{\sqrt{n}(\mu_{M_j}(F) - \widehat{\mu}_{j,n})}{S_{j,n}} \leq x\right\}. \tag{13}$$

However, in order to find $M_n(\beta)$, we need first find $\widehat{\mu}$ and $\widehat{\Sigma}$. In this paper, h_0 is assumed to be $E(Y|X)$, which allows general ways to estimate it such as kernel or series estimators. Given the consistent nonparametric estimator $\widehat{h} \equiv \widehat{h}_0 = \widehat{E(Y|X)}$, we could find the $\widehat{\mu}$ and $\widehat{\Sigma}$ by a reparameterized GMM condition (15) with the method provided by Newey (1994a). Thereby consider the reparameterized moment functions under H_0

$$g(W, \mu_M, h_0) \equiv m(W, \theta, h_0) - E[m(W, \theta, h_0)] = m(W, \theta, h_0) - \mu_M, \tag{14}$$

where

$$E[g(W, \mu_M, h_0)] = 0 \tag{15}$$

holds trivially. Then

$$\widehat{\mu} = \frac{1}{n}\Sigma_{i=1}^n m(W_i, \theta, \widehat{h}) \tag{16}$$

which minimizes the squared error loss function

$$\widehat{Q}_n(\mu_M) \equiv [\frac{1}{n}\Sigma_{i=1}^n g(W_i, \mu_M, \widehat{h})]'\widehat{\Gamma}[\frac{1}{n}\Sigma_{i=1}^n g(W_i, \mu_M, \widehat{h})],$$

where a weighting matrix $\widehat{\Gamma}\xrightarrow{p}\Gamma$ along with the identification of μ_M are assumed by Assumption 7 in the appendix A. Then by the Theorem 2.1 of Newey (1994a), we could find the influence function of $\widehat{\mu}$ under F by the pathwise derivative $d(W)$ along a path of $F_\tau \in \mathbf{F}$, a.k.a. a one-dimensional subfamily of $\mathbf{F}$, under some regularity conditions (RC) described in the appendix, so that the consistent estimator $\widehat{\Sigma}$ can be found as follows. Let $\| f \|$ denote the $\mathcal{L}^2$ norm for any generic function f.

Lemma 1. *Under RC as displayed in the appendix, if (i)* h_0 *is point identified and* $\| \widehat{h} - h_0 \| \xrightarrow{p} 0$*, (ii)* $\frac{1}{n}\Sigma_{i=1}^n \| \widehat{\alpha}(W_i) - \alpha(W_i) \|^2 \xrightarrow{p} 0$*, where* $\widehat{\alpha_j}(W_i) = \widehat{E\big(\frac{\partial m_j(W,\theta_0,h)}{\partial h}\big|_{h=h_0(x)}\big|X = x_i\big)}(y_i - \widehat{h}_0(x_i))$ *and* $\alpha_j(W_i) = E\big(\frac{\partial m_j(W,\theta_0,h)}{\partial h}\big|_{h=h_0(x)}\big|X = x_i\big)(y_i - h_0(x_i))$*, (iii)* $\| g(W,\mu,h) - g(W,\mu_M,h_0) \| \leq b(W)(\| \mu - \mu_M \| + \| h - h_0 \|)$ *while* $E[b^2(W)] < \infty$*, then*

$$\widehat{\Sigma}\xrightarrow{p}\Sigma, \tag{17}$$

where

$$\widehat{\Sigma} = \frac{1}{n}\Sigma_{i=1}^n \widehat{\iota}\times\widehat{\iota}' \tag{18}$$
$$\widehat{\iota_j} \equiv \widehat{\iota_j}(W_i,\theta_0,\widehat{h_0}(x_i)) = m_j(W_i,\theta_0,\widehat{h_0}(x_i)) - \frac{1}{n}\Sigma_{i=1}^n m_j(W_i,\theta_0,\widehat{h_0}(x_i)) + \widehat{\alpha_j}(W_i)$$

and

$$\Sigma = \mathbb{E}(\iota\times\iota') \tag{19}$$
$$\iota_j \equiv \iota_j(W,\theta_0,h_0) = m_j(W,\theta_0,h_0(x)) - \mu_{M_j} + \alpha_j(W_i)$$

Proof. First observe that $E[g(W,\mu_M,h_0)] = 0$ is the GMM moment condition, where g is defined as (14). We fix a θ_0 value at θ under H_0 so that only μ_M and h_0 are the unknown parameters. Then under the point-identification of μ_M and h_0 , the moment function g has randomness coming from $W\sim F$ only while the two parameters μ_M and h_0 will be determined once the DGP F is given. Then by Theorem 2.1 of Newey (1994a), under Assumption 2, there exists a pathwise derivative of μ_M denoted as $d(W)$ that equals to the influence function of $\widehat{\mu}$ denoted as $\psi(W)$ such that

$$\sqrt{n}(\widehat{\mu}_M - \mu_M) = n^{-\frac{1}{2}}\Sigma_{i=1}^n \psi(W_i) + op(1). \tag{20}$$

Under RC Assumption 3 and 5 that guarantees the functional $E[g(W,\mu_M,h_\tau)]$ is continuous and linear so that Riesz representation theorem is applicable , by Propositions 1, 4 and 5 of Newey (1994a),

$$\psi(W) = -G^{-1}[g(W,\mu_M,h_0) + \alpha(W)], \tag{21}$$

a $k \times 1$ vector, where $G \equiv \frac{\partial E[g(W,\mu,h_0)]}{\partial \mu}\big|_{\mu=\mu_M}$, $\alpha(W) = \delta(x)(y - h_0(x))$, $\delta(x) = E[\Delta(W)|x]$ and $\Delta(W) = \frac{\partial g(W,\mu_M,h)}{\partial h}\big|_{h=h_0(x)}$. Hence $G = -I_k$ by dominated convergence theorem given Assumption 1, where I_k is a k-dimensional identity matrix. Therefore, as $\Delta(W) = \frac{\partial}{\partial h} m(W_i,\theta_0,h)\big|_{h=h_0(x)}$,

$$\psi(W) = m(W,\theta_0,h_0) - \mu_M + E\Big(\frac{\partial m(W,\theta_0,h)}{\partial h}\big|_{h=h_0(x)}\Big|x\Big)(y - h_0(x)), \tag{22}$$

where $E(\psi(W)) = \mathbf{0}$. Hence, the asymptotic variance of $\widehat{\mu}_M$ is

$$\Sigma = var(\psi(W)) = E[\psi(W)\psi'(W)],$$

which is (19). The consistency result of the plug-in estimator $\widehat{\Sigma}$ is a direct consequence of Lemma 5.2 and Lemma 5.4 of Newey (1994a) under Assumption 3-8. □

Remark 1. $E\Big(\frac{\partial m_j(W,\theta_0,h)}{\partial h}\big|_{h=h_0(x)}\Big|X = x\Big)$ is the projection of $\frac{\partial m_j(W,\theta_0,h)}{\partial h}\big|_{h=h_0(x)}$ for $j = 1,\ldots,k$ on the plane of all square integrable functions of X. Its estimator $\widehat{E\Big(\frac{\partial m_j(W,\theta_0,h)}{\partial h}\big|_{h=h_0(x)}\Big|X = x\Big)}$ will be determined using a smooth function, such as kernel function or series function under some additional conditions for the nonparametric estimation. And the key feature of $\widehat{\Sigma}$, comparing to the normal one-step GMM is that we have an adjustment term regarding $\widehat{h}_0(x)$, which is

$$\widehat{E\Big(\frac{\partial m_j(W,\theta_0,h)}{\partial h}\big|_{h=h_0(x)}\Big|X = x\Big)}(y_i - \widehat{h}_0(x_i)),$$

$j = 1,\ldots,k$, in this paper. This adjustment term characterizes the effect of estimator $\widehat{h}_0$ on the asymptotic distribution of $\widehat{\mu}$ through the channel of the asymptotic normality of $\sqrt{n}E[g(\widehat{W,\mu_M},h_0)]$, where $E[g(\widehat{W,\mu_M},h_0)] \equiv \frac{1}{n}\Sigma_{i=1}^n m(W_i,\theta,\widehat{h}) - \mu_M$. Through $\widehat{\mu}$, which further affect $\widehat{\Sigma}$, along with $\widehat{\Sigma}$, the confidence sets in two steps, i.e. (12) and (23), will be characterized under bootstrapping. Apparently, if

the choice of $\widehat{\mu}$ and $\widehat{\Sigma}$ is problematic, the validity of the confidence set of parameters of interest will fail. If $\alpha(W) = 0$, $\widehat{h}_0$ will have no impact on it.

The estimator of $\alpha(W)$ could be formed by plug-in without knowing the actual function form of $\alpha(W)$ which could be shown immediately follows the proof of Lemma 4 in the Appendix B. The corresponding construction for kernel estimator for $\widehat{h}$ has been discussed in Newey (1994b) and the series estimators has been discussed in Newey (1994a).

Remark 2. For the case of N nuisance functions under

$$E[g(W, \mu_M, h_{10}, \ldots, h_{N0})] = 0,$$

the corresponding result in Lemma 1 remains with the following formula for the pathwise derivative.

$$\psi(W) = -G^{-1}[g(W, \mu_M, h_{10}, \ldots, h_{N0}) + \alpha(W)],$$

where

$$G \equiv \frac{\partial E[g(W, \mu, h_{10}, \ldots, h_{N0})]}{\partial \mu}\Big|_{\mu=\mu_M},$$

$$\alpha(W) = \alpha_1(W) + \cdots + \alpha_N(W),$$

$$\alpha_\nu(W) = E\Big(\frac{\partial g(W, \mu_M, h_1, \ldots, h_N)}{\partial h_\nu}\Big|_{h_\nu = h_{\nu 0}(x)}\Big| X = x_i\Big)(y_{\nu i} - h_{\nu 0}(x_i)),$$

and

$$h_{\nu 0}(x) = E(Y_\nu | X = x)$$

for $\nu = 1, \ldots, N$.

The second confidence set is then defined by

$$CS_n(\alpha - \beta) = \big\{\theta \in \Theta : T_n(\theta, \widehat{h}_0) \leq C_n(1 - \widehat{\alpha + \beta}, \theta, \widehat{h}_0)\big\}, \tag{23}$$

where

$$C_n(1 - \widehat{\alpha + \beta}, \theta, \widehat{h}_0) = \sup_{\lambda \in M_n(\beta) \bigcap \mathbb{R}^k_+} J_n^{-1}(1 - \alpha + \beta, \lambda, \widehat{F_n}),$$

and

$$J_n(x,\lambda,F) = P\Big(S\big(\widehat{D}^{-1/2}(\sqrt{n}(\widehat{\mu}-\mu(F))) + \widehat{D}^{-1/2}\sqrt{n}\lambda, \widehat{\Omega}_n\big) \leq x\Big) \tag{24}$$

represents the distribution of $T_n^{TS}(\theta)$ while $J_n(x,\mu,F) \equiv J_n(x,F)$ represents the distribution of $T_n(\theta)$. Under the null,

$$\lambda_j^* = max(\widehat{\mu}_j + \frac{S_j K_n^{-1}(\beta, \widehat{F_n})}{\sqrt{n}}, 0) \tag{25}$$

and

$$\widehat{C_n(1-\alpha+\beta,\theta,\widehat{h}_0)} = J_n^{-1}(1-\alpha+\beta,\lambda^*,\widehat{F_n}).$$

Then our task will be applying $T_n(\theta)$ and its approximated critical value $\widehat{C_n(1-\alpha+\beta,\theta)}$ from bootstrapping, ignoring $\widehat{h}$ hereafter, at level of α and discuss the UASC, which leads to the following main result. The algorithm of the two-step procedure with nuisance functions will be discussed in the next section.

Theorem 1. *Suppose $W_i, i = 1,\dots n$, is an i.i.d. sequence of random vectors with distribution $F \in \mathbf{F}$ and $\mathbf{F}$ satisfies*

$$\lim_{\lambda\to\infty} \sup_{(\theta,h,F)\in\mathcal{F}} \mathbb{E}_F\Big[\Big(\frac{\psi_j(W)}{\sigma_j(F)}\Big)^2 \mathbb{1}\Big\{\Big|\frac{\psi_j(W)}{\sigma_j(F)}\Big| > \lambda\Big\}\Big] = 0 \tag{26}$$

for all $1 \leq j \leq k$, where $\psi(W)$ is defined by (22). If RC as displayed in the appendix and the assumptions of Lemma 1 satisfied, then the test defined by the critical function

$$\phi_n \equiv \phi_n(\alpha,\beta) = 1 - \mathbb{1}\big\{\{M_n(\beta) \subseteq \mathbb{R}_+^k\} \cup \{T_n(\theta) \leq \widehat{C_n(1-\alpha+\beta,\theta)}\}\big\} \tag{27}$$

of H_0 satisfies the UASC property as defined by (9) for statistics (4), (6) and(7).

Remark 3. The uniform integrability condition requires uniformity over DGP, while the normal asymptotic tools such as the weak LLN and Lindeberg-Levy CLT only require for a fixed DGP. The UASC in this paper is achieved by applying Lindeberg CLT, while Andrews and Soares (2010a) apply Lyapunov CLT. All other RC assumptions concerning the DGP are regular for the sake of the approximation of the infinite dimensional parameter h_0 and the finite dimensional parameter μ_M. Therefore, the space of DGP $\mathcal{F}$ is sufficiently large.

Remark 4. The test (27) takes into account that the estimator $\widehat{h_0}$ of h_0 could take any value in the neighbor of h_0 including the case $E_{F_0}[m_j(W_i,\theta_0,h_0)] \geq 0, j = 1\dots k$ would be rejected as long as $\widehat{h_0}$ converges in probability to h_0 under an appropriate rate. That is, we could tolerate the violation of the nulls $H_{0'}$ and H_0, i.e. there exists a $j \in 1,\dots,k$ such that$E_{F_0}[m_j(W_i,\theta_0,h_0)] < 0$, but the frequency of the violation cannot be more than the levels defined in each tests. Therefore, the confidence set CS_n of interest allows to some extend the sample noises overall. However, in order to uniformly control the false positives including this, we need to introduce the Bonferroni-type correction for the two-step procedure amounts to Romano et al. (2014).

The proof of this theorem demands the asymptotic distributions of the test statistics and thereby relys on the uniform bootstrapping consistency in the presence of nuisance functions, motivated by the Theorem 2.4 of Romano and Shaikh (2012a). To save space, we provide the proof in the appendix.

3 Algorithm and Simulation Study

3.1 Algorithm for Partially Identified Parameters of Interest

(1) Find nonparametric estimator $\widehat{h} = \widehat{E(Y|X)}$ and $E\big(\frac{\partial m_j(\widehat{W,\theta_0,h_0(x)})}{\partial h_0(x)}|x\big)$ $\forall 1 \leq j \leq k$ by a kernel function or series function using the original sample;

(2) Find $\widehat{\mu}$ by (16) and $\widehat{\Sigma}$ by (18);

(3) Find T_n w.r.t. H_0 and the bootstrap critical value $K_n^{-1}(\beta,\hat{F}_n)$ derived by (13);

(4) Use $\sqrt{n}\lambda_j^*$ by (25) to replace $\sqrt{n}\mu_M$ in (8);

(5) Find the bootstrap critical value $\widehat{C_n(1-\alpha+\beta,\theta,\widehat{h}_0)} = J_n^{-1}(1-\alpha+\beta,\lambda^*,\widehat{F_n})$ derived by (24);

(6) Find the set of values of θ as defined by the confidence set (23).

During the process, we implicitly utilize the first step confidence set $M_n(\beta)$ through its lower left cortex value which will be regarded as the value of λ_n to replace μ_M as described by (25) so as to find the critical value $\widehat{C_n(1-\alpha+\beta,\theta,\widehat{h}_0)}$ of second step. One can fix $\beta = 0.005$ w.r.t. $\alpha = 0.05$ for simplicity. The confidence set (23) can be obtained by grid search or some other algorithms subject to the

moment inequality model therein. If applying the grid search, the step (6) will be repeating step (1) to (5) for all θ values in Θ to find the confidence set of interest.

The bootstrapping statistics is defined as followed for $T_n^{min} = \underset{1\le j\le k}{min} \frac{\sqrt{n}(\mu_{M_j}(F)-\widehat{\mu}_j)}{S_j}$ along with $T_n = \underset{1\le j\le k}{max} - \frac{\sqrt{n}\widehat{\mu_j}}{S_j}$ for the two hypothesis tests as defined by $H_{0'}$ and H_0 correspondently with $R = (R_1, R_2)$ repetitions for two separate bootstrapping resampling procedures. As shown in Lemma 7,

$$\underset{n\to\infty}{liminf}\underset{F\in\mathbf{F}}{inf} P\{J_n^{-1}(\alpha_1, \widehat{F_n}) \le T_n^{min} \le J_n^{-1}(1-\alpha_2, \widehat{F_n})\} \ge 1-\alpha_1-\alpha_2$$

which enables us to construct the critical value of the distribution of T_n^{min} behaving uniformly over $\mathbf{F}$. Therefore, the bootstrapping statistics for $K_n^{-1}(\beta, \hat{F}_n)$ under R_1 repetitions will be

$$\underset{1\le j\le k}{min} \frac{\sqrt{n}(\widehat{\mu}_j - \widehat{\mu^*}_j)}{S_j^*} \tag{28}$$

where $\widehat{\mu_j^*}$ and S_j^* denote the bootstrap sample mean and the bootstrap sample variance respectively generated by the nonparametric i.i.d. bootstrapping samples $\{W_{i,r}^* : i = 1, \ldots, n\}$ for $r = 1, \ldots, R_1$, each with sample size n, which are resampled from the original sample. Then by Lemma 2, $\frac{\sqrt{n}(\widehat{\mu_j}-\mu_{M_j}(F_n))}{\sigma_j(F_n)} \xrightarrow{d} \mathcal{N}(0,1)$, $\forall j$. Thus, under the Theorem 1, the bootstrapping statistics for $\widehat{C_n(1-\alpha+\beta, \theta, \widehat{h}_0)}$ under R_2 repetitions will be

$$\underset{1\le j\le k}{max}\left\{\frac{\sqrt{n}(\widehat{\mu}_j - \widehat{\mu_j^*})}{S_j^*} - \frac{\sqrt{n}\lambda_j^*}{S_j^*}\right\} \tag{29}$$

where $\widehat{\mu_j^*}$ and S_j^* are defined same as above.

3.2 Simulation Study

This simulation example is based on Andrews and Soares (2010b). Consider an interval outcome regression model based on a partial linear model

$$Y = \theta X + h(Z) + U$$

under the assumption of zero conditional mean

$$E(U|X,Z)=0.$$

Then

$$E[X^2(Y-\theta X-h(Z)]=0$$

and

$$E[\theta X+h(Z)-Y]=0.$$

Random variable X and Z are observable, however Y is not. Therefore, if there is no further assumptions or conditions, θ is not identified. For unobservable Y, we could instead observe an interval $Y_l \leq Y \leq Y_u$, where Y_l is the integer part of Y and Y_u is the smallest integer greater or equal to Y. Thus,

$$E(X^2Y_l) \leq E(X^2Y) \leq E(X^2Y_u).$$

Then we could form the moment inequalities

$$\begin{aligned} E[X^2(Y_u-\theta X-h(Z))] &\geq 0 \\ E[\theta X+h(Z)-Y_l] &\geq 0 \end{aligned} \tag{30}$$

where $m_1 \equiv X^2(Y_u-\theta X-h(Z))$ and $m_2 \equiv \theta X+h(Z)-Y_l$. The data available to econometricians are $\{X,Z,Y_l,Y_u\}$.

In this study, we specify that

$$h(Z)=E(X|Z),$$

$(X,Z) \sim \mathcal{N}\Big(\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}1 & 0.2\\ 0.2 & 1\end{pmatrix}\Big)$, $U \sim \mathcal{N}(0,1)$ and $\theta=1$. Then $h(Z)=0.2Z+0.8$. We will apply $T_n^{min} = \underset{1\leq j\leq k}{min} \frac{\sqrt{n}(\mu_{M_j}(F)-\widehat{\mu}_j)}{S_j}$ along with $T_n = \underset{1\leq j\leq k}{max} -\frac{\sqrt{n}\widehat{\mu_j}}{S_j}$ to find the finite sample coverage probability (CP) of the model above at level $\alpha=0.05$. And we fix $\beta = 0.005$ for simplicity. The CP is defined as the relative frequency of coverage of the true parameter θ. We will report the CP for sample size $n=100$, 500 and 2000.

The plan of the simulation includes three parts: (a) find the identified set of θ

using an independent sample of size $N = 1,000,000$; (b) generate a new sample and follow the step (1)-(5) of the algorithm described in the last section to determine the binary choice $B \equiv \mathbb{1}\{T_n \leq C_n(1-\widehat{\alpha+\beta},\theta,\widehat{h}_0)\}$ while $R_1 = R_2 = 1000$; (c) repeat (b) $R' = 20,000$ times to get a collection of value of binary choices $\{B_i\}_{i=1}^{R'}$ and calculate the $CP = \frac{\Sigma_{i=1}^{R'} B_i}{R'}$.

The identified set found in this case is $[0.4998, 1.2486]$. We will then simulate for θ' taking values of 0.4498, 0.4998, 1, 1.2486 and 1.3735, which include the cases of the value of θ outside the identified set, on the boundary of the identified set and the interior of the identified set. We should anticipate that the CP will be lower and lower for θ' outside the identified set and far from the boundary of the identified set while all those CPs will be less than 0.95. For all the CPs with respect to the θ' inside the identified set, it will be bigger or equal to 0.95. Particularly speaking, for all θ' taking interior values of the identified set, the CPs will tend to 1. And following the UASC property we derive, the CPs for θ' on the boundary of the identified set will tend to 0.95 as sample size increases. The following table displays the simulation results.

Table 1: Finite Sample Coverage Probabilities of Confidence Sets for θ Values with $\beta = 0.005$

θ	0.4498 (Out)	0.4998 (On)	1.0000 (In)	1.2486 (On)	1.3735 (Out)
$n = 100$	0.9199	0.9698	1.0000	0.9325	0.6675
$n = 500$	0.7575	0.9539	1.0000	0.9517	0.1952
$n = 1000$	0.6261	0.9585	1.0000	0.9578	0.0274

Note.—The Nominal Level is $\alpha = 0.05$.

According to the results, all the anticipations have been verified, which is consistent with the UASC result we have shown in the Theorem 1. We also conduct the simulation for $\theta = 0.4498$ when $n = 6000$, the coverage probability is 0.0438. Thus, although the finite sample power might not be big enough, the asymptotic power will tend to 1.

In order to motivate the idea of choosing β close to zero, we implement another simulation for $\beta = 0.0001$ while keeping the same random number generator. In addition to all the results discussed above hold, when β is chosen a smaller

one, the performance is even better. In particular, the CPs for θ' on the boundary of the identified set become closer to 0.95 and the CPs for θ' outside the identified set will be closer to zero. The following table displays the simulation results. Based on the results, for small β, the test would not be overly conservative.

Table 2: Finite Sample Coverage Probabilities of Confidence Sets for θ Values with $\beta = 0.0001$

θ	0.4498 (Out)	0.4998 (On)	1.0000 (In)	1.2486 (On)	1.3735 (Out)
$n = 100$	0.9118	0.9665	1.0000	0.9503	0.7543
$n = 500$	0.7433	0.9485	1.0000	0.9484	0.1853
$n = 1000$	0.6089	0.9537	1.0000	0.9532	0.0254

Note.—The Nominal Level is $\alpha = 0.05$.

4 Empirical Study

When there are parameters of interest involved in strategic behaviors of economic agents, one can find the economic equilibrium upon the information structure with a specified solution concept and adopt this economic equilibrium to form a prediction model in terms of Conditional Choice Probability (CCP). And the econometric model in terms of moment conditions can then be established via appropriate transformation of the prediction model so that one can apply certain method to infer the parameters of interest.

A typical general decision making process is that the decision makers first form their updated beliefs about opponents' private signals while there are asymmetric information of private signals among all players and then make the corresponding decisions which leads to certain outcomes. A usual solution concept is Bayesian Nash Equilibrium (BNE), which is more general than Nash Equilibrium. The outside observer econometricians usually have asymmetric information from all the players in terms of the private signals and the parameters in the payoff functions of players, etc. The econometricians could then regard the private signals as random shocks to build up the equality between the true CCP, representing the true probability of choices made by players, and the predicted choice probability (PCP), measuring the events of random shocks that could predict any choices as the selected equilibrium choices, as the prediction model.

Then either the conditional or unconditional moment conditions on the basis of the prediction model can be found. Based on the dataset obtained and the reasonable assumptions made, the parameters of interest could then be either point identified or partially identified.

Under this procedure, the function form of the CCP is usually unknown to econometricians and it will be regarded as a nuisance function. And in general, CCP exists widely when we study the behavior of individual economic agents. In addition, the nuisance functions can also occur to econometrician when the function form of the payoff function, a function of covariates, is unknown to econometrician, even if known to all players. The function form of PCP, on the other hand, given the specified economic structure, might be known to econometrician. Therefore, addressing with the nuisance functions while conducting inference on the parameters of interest is of a general interest in the empirical work, especially under a structural econometric model.

4.1 The Structure Model

In this section, we illustrate the two-step method with nuisance function (NTS) by a static interaction game with incomplete information, motivated by De Paula and Tang (2012a) which addresses with the point identification of the sign of the state-dependent interaction effect. In our application, we are instead interested in the actual values of the state-dependent interaction effect, though partially identified, under a different specification of private signals. We want to know whether the radio stations have incentives to coordinate the commercial breaks upon imperfect monitor from the advertisers, that can potentially increase the listenership of the advertisement.

There are N players each player i with a choice set of two elements $\mathcal{D}_i \equiv \{0,1\}$. The payoff functions of player i are

$$\begin{aligned} \pi_{1i} &= \xi_i' X + \delta_i' X \sum_{j \neq i} D_j + \epsilon_i, \quad \text{if } D_i = 1 \\ \pi_{0i} &= 0, \quad \text{if } D_i = 0 \end{aligned} \tag{31}$$

where $\epsilon_i | X \sim \mathcal{U}[l,u]$ and $i \in \mathcal{I} \equiv \{1,\ldots,N\}$. The information structure is supposed as follows: $\xi_i' X$ and $\delta_i' X$ are common knowledge among all players, player

i privately observe a signal ϵ_i, whose distribution is a common prior among all players, player i will make his own choice D_i but does not know what the opponents $(-i)$ will choose; we outside observer econometrician have access to the covariates X and the strategy profiles taken by the players, i.e. $D \equiv \{D_i\}_{i\in\mathcal{I}}$, besides we also know $\epsilon_i|X \sim \mathcal{U}[l,u]$ but does not know any realized value of ϵ_i. The dataset thereby consists of G cross-sectional games each $g \in \mathcal{G} \equiv\{1,\ldots,G\}$ with their corresponding values of X and D. We assume rationality of all players throughout the games. We are interested in learning the parameters $\theta_0 \equiv \{\xi_i,\delta_i\}_{i\in\mathcal{I}}$. However, like most of the empirical datasets which usually have limited variations in covariates and thereby the standard assumptions for identification such as the rich support proposed in Tamer (2003) and Grieco (2014) is hard to be satisfied in many cases, we focus on the partial identification of parameters of interest while the NTS will work.

The solution concept adopted is pure strategy Bayesian Nash Equilibrium (BNE). The pure strategy BNE exists for any given X, which can be proved by the Theorem 1 of Athey (2001) applying Kakutani's fixed point theorem where the key is that the payoff functions satisfy the single crossing condition[3] while the distribution of $\epsilon|X$ is atomless. Meanwhile, the pure strategy BNE D_i is nondecreasing in private signal ϵ_i for all i. Therefore, conditional on X, the equilibrium condition for player i is

$$D_i(X,\epsilon_i) = \begin{cases} 1, & \text{if } \xi_i'X + \delta_i'X\sum_{j\neq i} E[D_j(X,\epsilon_j)|X,\epsilon_i] + \epsilon_i \geq 0; \\ 0, & \text{otherwise.} \end{cases}$$

We assume the conditional independence, that is

$$F_{\epsilon|X}(\cdot|x) = \Pi_{i\leq N} F_{\epsilon_i|X}(\cdot|x) \tag{32}$$

for any $x \in \Omega_X$. Then

$$E[D_j(X,\epsilon_j)|X=x,\epsilon_i] = E[D_j(X,\epsilon_j)|X=x] \equiv p_j(x).$$

This implies a cutoff rule $\kappa(x)$ for best responses to all players and player i will

[3] In particular, "$D_{Hi} \equiv 1 > D_{Li} \equiv 0$, $\epsilon_{Hi} > \epsilon_{Li}$, $\pi_i(D_{Hi},\epsilon_{Li}) - \pi_i(D_{Li},\epsilon_{Li}) \geq 0$" implies "$\pi_i(D_{Hi},\epsilon_{Hi}) - \pi_i(D_{Li},\epsilon_{Hi}) \geq 0$".

play 1 if

$$\epsilon_i \geq \kappa_i(x) \equiv -\xi_i' x - \delta_i' x \sum_{j \neq i} p_j(x)$$

while the function form of $p_j(\cdot)$ is fixed at equilibrium. At the equilibrium threshold $\kappa_i^*(x) \equiv \epsilon_i^*(x)$, player i is indifferent from playing 0 and 1, that is

$$\epsilon_i^*(x) = -\xi_i' x - \delta_i' x \sum_{j \neq i} p_j^*(x) = -\xi_i' x - \delta_i' x \sum_{j \neq i} \frac{u - \epsilon_j^*(x)}{u - l}.$$

Then

$$\epsilon_i^*(x) = \frac{-a_i(x)(u-l) - b_i(x)(N-1)u}{u - l + b_i(x)} + \frac{b_i(x)}{u - l + b_i(x)} \times \frac{A(x)}{1 - B(x)}, \tag{33}$$

where $a_i(x) = \xi_i' x$, $b_i(x) = \delta_i' x$, $A(x) = \sum_{i=1}^N \frac{-a_i(x)(u-l) - b_i(x)(N-1)u}{u-l+b_i(x)}$ and $B(x) = \sum_{i=1}^N \frac{b_i(x)}{u-l+b_i(x)}$ for all i, which defines the unique pure strategy BNE, i.e. player i will play 1 if observe $\epsilon_i \geq \epsilon_i^*(x)$ and play 0 otherwise.

Note that, in addition to $\kappa(x) \equiv (\kappa_1(x), \ldots, \kappa_N(x))'$, the BNE can also be expressed in terms of CCP , i.e. $p(x) \equiv (p_1(x), \ldots, p_N(x))'$. It implies that the strategy profile is a multivariate Bernoulli random vector where each components are conditionally independent. The correspondence between $\kappa(x)$ and $p(x)$ is bijective conditional on X when $F_{\epsilon_i|X}(\cdot|x)$ is invertible on its support. Thus, the BNE is equivalently characterized by

$$p_i^*(x) = E[\mathbf{1}\{\epsilon_i \geq \epsilon_i^*(x)\}|X = x] = \frac{u - \epsilon_i^*(x)}{u - l} \tag{34}$$

for any i.

The prediction model applying the BNE is thereby

$$H(\theta, X) \leq p(X) \leq H(\theta, X), \tag{35}$$

where $H(\theta, X) = [H_1(\theta, X), \ldots, H_N(\theta, X)]'$, $H_i(\theta, X) = \frac{u - \epsilon_i^*(X)}{u-l}$ and $p_i(X) = E[D_i|X]$ for any i . The model is apparently coherent and complete due to the uniqueness of BNE.

There is no equilibrium selection mechanism (ESM) involved due to uniqueness of BNE. One may change the primitive of distribution of $\epsilon|X$ to other atom-

less distributions which may result in multiple equilibriums and ESM will then naturally enter the model. But that is just an conceptually extension with more computation burden and is not the focus of this application. Besides, the misspecified ESM might lead to significant bias that has been discussed by Honor and De Paula (2010).

The usual ways to deal with the multiplicity of equilibriums have been discussed in Ciliberto and Tamer (2009), Beresteanu et al. (2011) and Grieco (2014). The first way is extending the prediction model (35) to the inequality of CCP for the counterfactual equilibrium strategy profiles while the lower bound is the probability that the corresponding counterfactual equilibrium is the unique equilibrium and the upper bound is the maximal probability that the corresponding counterfactual equilibrium is one of equilibriums.[4] The ESM is involved but we can grid search θ without find it given least restrictions imposed on the ESM. The second way is extending (35) to an equality $p(X) = H(\theta, X, \varrho)$ where ϱ is the ESM and can be found along with θ. Besides, the identification and estimation of ESM has also been discussed in Sweeting (2009) and Bajari et al. (2010) under parametrization of ESM.

The econometric model or the moment inequality model is then

$$E[H(\theta, X)] \leq h \leq E[H(\theta, X)], \tag{36}$$

which we can typically apply the NTS to get the confidence set of θ as defined by (2) to partially identify θ with the nuisance function $h \equiv p(X)$ therein.

We take a necessary condition of (35) for a collection of observations of X from different games rather than a particular observation of X. One could further consider conditional moment inequality model as discussed by Andrews and Shi (2013), Chernozhukov et al. (2013), Andrews et al. (2023) and Cox and Shi (2023), but it is not the focus of this study. Ciliberto and Tamer (2009) has a similar structure model as the form of (36), however, they use a simple frequency estimator to approximate the nuisance function h without fully consider the approximation error that would have significant impact on the confidence set of interest, through the channel of the estimator of the asymptotic variance of the sample mean of the corresponding moment functions as discussed in section 2.

[4] Because the ESM in the PCP is bounded between 0 and 1.

In the above model, $\xi_i' X$ is regarded as the basic payoff at state X for player i. $\delta_i' X \sum_{j \neq i} D_j$ captures the homogeneous state-dependent interaction effect at state X upon what the opponents do to player i and also explains how the state variables will affect the payoff of a player. One can also regard the basic payoff as nuisance function $u(X)$ as long as the partial derivative $\partial m / \partial u$ exists. One can extend to the case with heterogeneous state-dependent interaction effect using our method but it is not the focus of this paper. $\epsilon \equiv \{\epsilon_i\}_{i \in \mathcal{I}}$ as private signals are the only random shocks to econometrician. One can extend to the case when public shocks known to all players but unknown to econometricians play a role with our method if the conditional joint distribution of public shocks and private signals is given to econometricians. Besides, one can also extend to the case where the private signals are correlated or follow a conditional joint distribution known up to finite dimensional parameters, as long as it is atomless. Our method can also be applied to other type of games with different solution concepts, such as airline entry games discussed in Ciliberto and Tamer (2009), voting communication games discussed in Iaryczower et al. (2018) and marriage matching games discussed in Gualdani and Sinha (2023). In particular, it can be applied to general support of D as long as the payoff functions satisfy the single-crossing condition proposed by Athey (2001).

Regarding Mixed Strategy Nash Equilibriums (MSNE) as discussed in Berry and Tamer (2006) and Beresteanu et al. (2011), one need to further consider the randomization among pure strategies mixed in MSNE, which is a simple conceptually extension with more computation burden. One could also consider the model selection between two moment inequality models implied by different information structure by a hypothesis testing procedure as discussed in Shi (2015). A general framework of forming moment inequality models under certain conditions or assumptions within a microeconomic theory structure has been discussed in Pakes et al. (2015) and a general discussion of applying moment inequality models in different microeconomic context can be found in Molinari (2020).

Compared to De Paula and Tang (2012a), we share the same assumption of conditional independence. However, they assume unbounded support of the private signals, while we specify the bounded support. Another key compared feature of this structure model is that the ESM does not get involved at all, i.e.

the equilibrium is unique, while specifying the conditional distributions of private signals. Though De Paula and Tang (2012a) is applicable to any nonparametric ESM, our method works for any suitable distributions if not misspecified. Moreover, the potential inconsistency issue that the true CCP does not match the theoretical CCP[5] and the true CCP does not satisfy the equilibrium conditions are avoided. In addition, the prediction bases on the equilibrium solution rather than equilibrium conditions. The rationale is that under our model the unique BNE is achieved and for any observed choice data, players are supposed to make their best response by directly choosing one pure strategy out of their choice set upon their private signals. And the variation of the data solely contribute to the partial identification by the confidence set of interest, without the concern of misspecification of ESM[6].

Our model completes De Paula and Tang (2012a) by resolving the potential partial identification issue, especially when the covariances of equilibrium strategies of players and their opponents are zero. Meanwhile, De Paula and Tang (2012b) discusses other cases of nonidentification of the full structure, that further motivates our method. Besides, our model is capable to deal with continuous covariate X while De Paula and Tang (2012a) is limited to discrete X.

4.2 The Confidence Set of Interaction Effect

In this section, we apply the moment inequality model and discuss the confidence set of the interaction effect via test statistics (4) and (10) accordingly.

We use the dataset from Sweeting (2009)[7] to investigate the interaction effect represented by $\{\delta_i\}_{i=1,\ldots,N}$ of three radio stations, i.e. $N = 3$, who make one-shot decisions about whether entering the commercial break markets or not among their programming schedules denoted by D_i. The player i choose to enter the corresponding market at :55 of any hours if $D_i = 1$. The decisions are made on

[5]The theoretical CCP is the structural equilibrium CCP. The true CCP is the mixture of theoretical CCPs based on the ESM. When there is no ESM, the theoretical CCP meets the true CCP.

[6]ESM reflects the strategic behavior of economic agents. When finding the estimators of parameters of interest, the misspecified ESM might be chosen or estimated just to minimize the distance between CCP and PCP.

[7]We appreciate Andrew Sweeting for providing the data and helpful suggestions. We also appreciate Xun Tang for providing the detailed data management clarifications and relevant codes. The dataset of De Paula and Tang (2012a) is derived from that of Sweeting (2009).

four hours, including 4-5pm and 5-6pm for drivetime and 12-1pm and 9-10pm for non-drivetime. Their decisions are state (X) dependent and private signal (ϵ_i) dependent. The state variable X is the normalized market size of each commercial break markets taking values from positive real numbers. The total number of cross-sectional games is 26,152. The dataset comes from BIAfn's MediaAccess Pro data base, Media base 24/7 and the 2001 Census. The detailed description of the data can be found in Sweeting (2009).

We assume the payoff functions π_{1i} and π_{0i} take the form as (31), $\epsilon_i|X \sim \mathcal{U}[-0.5, 0.5]$, conditional independence of ϵ_i for all i as (32), the information structure described above and rationality of all players. We formulate the prediction model based on the BNE (34). Assume all the assumptions in NTS are satisfied, the moment inequality model is as followed for $i \in \mathcal{I} \equiv \{1, 2, 3\}$.

$$E[h_i - H_i(\theta, X)] \geq 0$$
$$E[H_i(\theta, X) - h_i] \geq 0$$

In this model, there are six moment functions while two for each players. The parameters of interest are $\{\xi_i, \delta_i\}_{i\in\mathcal{I}}$ consisting of 6 parameters. In addition, there are three infinite-dimensional nuisance parameters $h_i = E(D_i|X)$ which can be estimated by series functions with intercept using GCV criteria.

For simplicity, we normalize the parameters by $\xi_i = 1$ since only the cutoff matters in the decision making process of radio stations. Besides, we assume homogeneity in δ_i, based on the symmetric choice pattern observed, and find the confidence set of $\delta_i \equiv \delta$, among $[-100, 100]$ with three decimal places as the parameter space, by grid search.

The confidence set of δ [8] for the whole data is

$$\{[-3.113, -2.973], [0.068, 6.686]\}$$

with 137 negative points and 299 positive points. For the drivetime markets, the confidence set is

$$\{[-2.99, -2.917], [0.068, 6.687]\}$$

[8] All the confidence sets base on $R_1 = R_2 = 1000$ while $\alpha = 0.05$ and $\beta = 0.005$.

with 71 negative points and 421 positive points. While for the small markets, it is

$$\{[-4.622, -4.363], [1.816, 6.684]\}$$

with 335 negative points and 237 positive points. That for drivetime small markets is

$$\{[-4.418, -4.213], [1.816, 6.685]\}$$

with 195 negative points and 324 positive points. The state-dependent interaction effect will be $\delta' X$ where X takes positive values only.

The results provide evidence for both negative and positive interaction effects across different types of markets, which is different from De Paula and Tang (2012a). It also shows a robust positive upper bound for the interaction effect. Another feature of the confidence set is the nonconvexity, which again motivates our method that applies to nonconvex identified set.

However, compared to the whole data, the grid points of interaction effect searched for the data of drivetime markets concentrate more on positive numbers with the negative interval shifting to the right and contracting. This concentration change towards positive interaction effect supports De Paula and Tang (2012a), which identifies the positive interaction effect in drivetime markets. Similar result is found in small markets during drivetime, compared to the pooled small markets.

To clarify the difference, this model discusses the interaction effect in the view of bounded support of $\epsilon_i|X$ while De Paula and Tang (2012a) does that of unbounded support. ϵ_i reflects the private benefit and cost to radio station i for each entry decisions of airing the commercials. Therefore, bounded support of the distribution of $\epsilon_i|X$ is reasonable. And the uniqueness of the BNE is the result of one specification of such distributions of $\epsilon_i|X$. However, the test of it as well as general identification and estimation of the distribution of $\epsilon_i|X$ for potential misspecification is of a great interest and is left to the future research.

One important implication of our structure model is that the uniqueness of equilibriums does not imply the point identification of the interaction effect. While the equilibriums depend on the information structure of the game, the identification of the interaction effect depends on the dataset and the identification assumptions if any imposed.

5 Conclusion

In this article, we provide a general method to conduct inference on finite-dimensional partially identified parameters of interest in finite number of unconditional moment inequalities with point identified infinite-dimensional nuisance parameters. We show that our test to form the confidence set of interest achieves the UASC through uniform bootstrapping consistency. We propose a structure model in game theory to illustrate the importance of our method in terms of CCP that is a nuisance function but has not been fully addressed with the effect of the approximation error of this nuisance function on the confidence set of interest. Besides, the method could be widely applied in economics as a general semiparametric model for partially identified parameters of interest.

The core way we characterize the effect of the estimator of nuisance parameters on the confidence set of interest is by a reparameterized GMM condition, that transfers the moment inequality model into a moment equality model to allow us to apply the pathwise derivative. The asymptotic variance of the sample mean of the moment functions is thereby derived.

The main caveat, however, is that the CAN property of the nuisance parameters is needed to hold that subject to further primitive conditions for the specific choice of approximation methods, albeit we assume it is point identified. And although our method is computationally feasible to some extent, the further reduction of computation burden is of a great interest in the future research. The general partial identification of conditional moment inequalities is yet also a open question to be discussed.

APPENDIX

Section A provides the Regularity Condition (RC) assumptions, section B provides the auxiliary lemmas and section C delivers the proof of the Theorem 1.

A RC assumptions

Assumption 1. $\mu_M < \infty$.

This technical assumption guarantees the exchange of orders of differentiation and integration.

Assumption 2. *(Pathwise Derivative) (i) the set of scores for regular paths is linear; (ii) for any* $\epsilon > 0$ *and measurable s(W) with* $E[s(W)] = 0$ *and* $E[s(W)^2) < \infty$*, there is a regular path with score* $S(W)$ *satisfying* $E[|s(W) - S(W)|^2] < \epsilon$*; (iii)* $\widehat{\mu}_M$ *is asymptotically linear and regular.*

This assumption guarantees the existence of the pathwise derivative of interest with the crucial requirement for asymptotic equivalence of (20). The definition of regular paths and regular asymptotic linear (RAL) estimators could be found as in Newey (1994a). What follows are the sufficient conditions for the CAN property of the estimator of μ_M or equivalently (20) in this paper.

Assumption 3. *(Linearization) (i) There is a function* $\Delta(W,h)$ *that is linear in h such that for all* h *with* $\| h - h_0 \| < \epsilon$ *for any* $\epsilon > 0$ *,* $\| g(W,\mu_M,h) - g(W,\mu_M,h_0) - \Delta(W,h-h_0) \| \leq b(W) \| h - h_0 \|^2$ *and* $E(b(W))\sqrt{n} \| \widehat{h} - h_0 \|^2 \xrightarrow{p} 0$.

This assumption guarantees the remainder term from linearization while applying Riesz representation theorem is small and the convergence rate of $\widehat{h}$ is faster than $n^{-\frac{1}{4}}$ when the moment function g is not linear.

Assumption 4. *(Stochastic Equicontinuity)* $n^{-\frac{1}{2}}\Sigma_{i=1}^n[(\Delta(W_i,\widehat{h}-h_0) - \int \Delta(W,\widehat{h}-h_0)dF_0] \xrightarrow{p} 0$.

This Assumption in terms of the above empirical process, not necessarily in the form of sample average, is the key assumption to be applied of the generic uniform law of large number as discussed in Newey (1991) and Andrews (1992). One typical sufficiency of stochastic equicontinuity is in terms of Lipschitz condition. The asymptotic normality of semiparametric estimators, or $\widehat{\mu}_M$ in this case, derived by stochastic equicontinuity has been discussed in Andrews (1994). This assumption along with the last one are requirements on second order terms which essentially requires the moment function $g(W,\mu_M,h)$ is sufficiently smooth and the estimator $\widehat{h}$ is well-behaved.

Assumption 5. *(Mean-square Continuity)(i) There is a function* $\alpha(W)$ *and a measure* $\hat{F}$ *such that* $E(\alpha(W)) = 0$*,* $E[\| \alpha(W) \|^2] < \infty$*, and for all* $\| \widehat{h} - h_0 \| < \epsilon$ *for any* $\epsilon > 0$*,* $\int \Delta(W,\widehat{h}-h_0)dF_0 = \int \alpha(W)d\hat{F}$*; (ii) For the empirical distribution* $\tilde{F}(w) = \frac{1}{n}\Sigma_{i=1}^n \mathbb{1}\{W_i \leq w\}$, $\sqrt{n}[\int \alpha(W)d\hat{F} - \int \alpha(W)d\tilde{F} \xrightarrow{p} 0$.

This smoothness type of assumption, which has be discussed more detailed in Newey (1990), is crucial to find the unique adjustment term $\alpha(W)$ and to derive the asymptotic normality of $n^{-\frac{1}{2}}\Sigma_{i=1}^{n} g(W_i, \mu_M, \widehat{h})$ given the consistency of $\widehat{h}$, which might converge in a different rate. It regards $\int \alpha(W) dF(W,h)$ that will be differentiable in h if $dF(W,h)^{1/2}$ has mean square a mean square derivative, which leads to the differentiability of $E[g(W, \mu_M, h_\tau)]$ in τ of a path $\{F_\tau\}$ in $\mathbf{F}$. It along with the linearization assumption makes the Riesz representation theorem applicable for the derivation of the influence function $\psi(W)$. The assumption 3-5 together guarantees the proper approximation of $\sqrt{n}\widehat{g}_n(\mu_M)$ by $\Sigma_{i=1}^{n}[g(W_i, \mu_M, h_0) + \alpha(W_i)]/\sqrt{n} + op(1)$ that leads to the asymptotic normality of $\sqrt{n}\widehat{g}_n(\mu_M)$ given the consistency of $\widehat{h}$ under a proper rate of convergence.

Assumption 6. *There are* $\epsilon, \| \ h \ \|, b(W), \tilde{b}(W) > 0$ *such that (i) for all* $\mu \in \mathcal{M}$, $g(W, \mu, h_0)$ *is continuous at* μ *with probability one,* $\| \ g(W, \mu, h_0) \ \| \leq \ b(W)$; *(ii)* $\| \ g(W, \mu, h) - g(W, \mu, h_0) \ \| \leq \tilde{b}(W)(\| \ h - h_0 \ \|)^{\epsilon}$.

This assumption provides an uniform Lipschitz condition of order ϵ for the uniform convergence of $\| \ \hat{g}_n(\mu) - E[g(W, \mu, h_0)] \ \|$ over $\mu \in \mathcal{M}$, i.e. $\underset{\mu \in \mathcal{M}}{sup} \| \ \hat{g}_n(\mu) - E[g(W, \mu, h_0)] \ \| \xrightarrow{p} 0$ similar to Andrews (1992).

Assumption 7. *(i) there exists an estimator* $\hat{\Gamma}$ *of a weighting matrix* Γ *such that* $\hat{\Gamma} \xrightarrow{p} \Gamma$, *where* Γ *is positive semi-definite; (ii)* $\Gamma E[g(W, \mu, h_0)] = 0$ *has a unique solution on compact* $\mathcal{M}$ *at* μ_M.

This assumption assures the identification of μ_M which then could deliver the consistency of $\widehat{\mu}$ combined with the last assumption.

Assumption 8. *(i)* $\mu \in interior(\mathcal{M})$; *(ii) there is* $\| \ h \ \|, \epsilon > 0$ *and a neighborhood* $\mathcal{N}$ *of* μ_M *such that for all* $\| \ h - h_0 \ \| < \epsilon$, $g(W, \mu, h)$ *is differentiable in* μ *on* $\mathcal{N}$; *(iii)* $G'\Gamma G$ *is nonsingular for* $G \equiv E\big[\frac{\partial g(W, \mu, h_0)}{\partial \mu}\big|_{\mu = \mu_M}\big]$; *(iv)* $E[\| \ g(W, \mu_M, h_0) \ \|^2] < \infty$; *(v) Assumption 6 is satisfied with* $g(W, \mu, h)$ *there equal to each row of* $\partial g(W, \mu, h)/\partial \mu$.

This assumption provides an uniform Lipschitz condition of order ϵ as Assumption 6 to deliver the consistency of the Jacobian

$$\frac{1}{n}\Sigma_{i=1}^{n} \nabla_\mu g(W_i, \tilde{\mu}, \widehat{h}) \xrightarrow{p} E[\nabla_\mu g(W, \mu_M, h_0)],$$

which leads to the asymptotic normality of $\widehat{\mu_M}$ and the consistency of $\widehat{\Sigma}$.

Remark 5. The RC Assumption 3-8 together could deliver the asymptotic normality of $\widehat{\mu_M}$ by mean-value expansion as discussed in Newey and McFadden (1994).

B Auxiliary Results

Lemma 2. *Let* W_i, $i = 1,\ldots,n$, *be an i.i.d. sequence of random vectors with distribution* $F_n \in \mathbf{F}$ *on* $\mathbb{R}^p$. *Suppose* $\mathbf{F}$ *is such that for all* $1 \leq j \leq k$, $\lim_{\lambda\to\infty}\sup_{F\in\mathbf{F}}\mathbb{E}_F\left[\left(\frac{\psi_j(W)}{\sigma_j(F)}\right)^2 \mathbb{1}\left\{\left|\frac{\psi_j(W)}{\sigma_j(F)}\right| > \lambda\right\}\right] = 0$. *If RC satisfied, then under* F_n, $\forall j$, $\frac{\sqrt{n}(\widehat{\mu_j}-\mu_{M_j}(F_n))}{\sigma_j(F_n)} \xrightarrow{d} \mathcal{N}(0,1)$.

Proof. By Lemma 1, $\sqrt{n}(\widehat{\mu}_M-\mu_M) = n^{-\frac{1}{2}}\Sigma_{i=1}^n\psi(W_i)+op(1)$ where $\psi(W_i)$ is characterized as (22) and $E(\psi(W_i)) = 0$. Since W_i is i.i.d., $\psi(W_i)$ is i.i.d.. Then by Lemma 11.4.1 of Romano and Lehmann (2005), $\frac{n^{-\frac{1}{2}}\Sigma_{i=1}^n\psi_j(W_i)}{\sigma_j(F_n)} \xrightarrow{d} \mathcal{N}(0,1)$. The desired result followed then by Slutsky's theorem. □

Remark 6. This lemma relies on that $\sigma_j(F_n)$ cannot tend to zero. There is a DGP of $\psi_j(W)$ on $\mathbb{R}$ uniquely determined by the DGP F of W so that Lemma 11.4.1 of Romano and Lehmann (2005), derived by Lindeberg CLT, is applicable. The asymptotic equivalence expansion is derived upon the true DGP F_0. Given the true DGP $F_0 = F_n$, we could have the above dynamic DGP CLT.

Lemma 3. *Let* $Y_{n,i},\ldots,Y_{n,n}$ *be i.i.d. with distribution* $G_n \in \tilde{\mathbf{F}}$ *with finite mean* $\mu(G_n)$ *and* $\tilde{\mathbf{F}}$ *satisfies*

$$\lim_{\lambda\to\infty}\sup_{F\in\tilde{\mathbf{F}}} E_F\left[\frac{|Y-\mu(F)|^2}{\sigma^2(F)}\mathbb{1}\{\{\left\{\frac{|Y-\mu(F)|}{\sigma(F)} > \lambda\right\}\right] = 0 \tag{37}$$

Let $\overline{Y}_n = \Sigma_{i=1}^n \frac{Y_{n,i}}{n}$. *Then under* G_n, $\overline{Y}_n - \mu(G_n) \xrightarrow{p} 0$.

This Lemma is similar to Lemma 11.4.2 of Romano and Lehmann (2005) but under a different assumption.

Proof. The proof follows the arguments of the proof of Lemma 11.4.2 of Romano and Lehmann (2005) by define $Z_{n,i} \equiv Y_{n,i}\mathbb{1}\{|Y_{n,i}| \leq n\}$ and $m_n \equiv E(Z_{n,i})$ so that we could prove the result by an intermediate result $\overline{Y_{n,i}} \xrightarrow{p} m_n$, which can be shown by two functions $\tau_n(t) \equiv t[1-G_n(t)+G_n(-t)]$

and $\kappa_n(t) \equiv \frac{1}{t}\int_{-t}^{t} x^2 dG_n(t)$ that transform it into showing $P(|\overline{Y_n} - m_n| > \epsilon) \leq \epsilon^{-2}\kappa_n(n) + \tau_n(n) \to 0$. WLOG, let $\mu(G_n) = 0$. By observing that: $\lim_{n\to\infty} E_{G_n}\big(|Y_{n,i}|^2 \mathbb{1}\{\{|Y_{n,i}| > n\}\}\big) = \lim_{n\to\infty} E_{G_n}\big(|Y_{n,i}|^2 \big| \{|Y_{n,i}| > n\}\big) \times P(|Y_{n,i}| > n) \geq \lim_{n\to\infty} E_{G_n}\big(|Y_{n,i}| \big| \{|Y_{n,i}| > n\}\big) \times P(|Y_{n,i}| > n) = \lim_{n\to\infty} E_{G_n}\big(|Y_{n,i}| \mathbb{1}\{\{|Y_{n,i}| > n\}\}\big)$,we will still have $\lim_{n\to\infty} \tau_n(n) = 0$ since $0 \leq \tau_n(n) = nP\big(|Y_{n,i}| > n\big) \leq E_{G_n}\big[|Y_{n,i}| \mathbb{1}\{\{|Y_{n,i}| > n\}\big]$ for any $n > 0$ and the uniform integrability condition (37) , which implies $\lim_{n\to\infty} E_{G_n}\big(|Y_{n,i}|^2 \mathbb{1}\{\{|Y_{n,i}| > n\}\}\big) = 0$, and $\kappa_n(n) \to 0$ because $\forall \delta > 0$, $\exists \beta_0$ such that $\underset{n\to\infty}{limsup} E[|Y_{n,i}| \times \mathbb{1}\{|Y_{n,i}| > \beta_0\}] < \frac{\delta}{4}$ while $n = \beta_0$. Then $m_n \xrightarrow{p} 0$ by the same observation. The desired result thereby is achieved by continuous mapping theorem. □

Lemma 4. *Let W_i, $i = 1, \ldots, n$, be an i.i.d. sequence of random vectors with distribution $F_n \in \mathbf{F}$ on $\mathbb{R}^p$. Suppose: (i) $\mathbf{F}$ is such that for all $1 \leq j \leq k$,*

$$\lim_{\lambda\to\infty} \underset{F\in\mathbf{F}}{sup} \mathbb{E}_F\Big[\Big(\frac{\psi_j(W)}{\sigma_j(F)}\Big)^2 \mathbb{1}\Big\{\Big|\frac{\psi_j(W)}{\sigma_j(F)}\Big| > \lambda\Big\}\Big] = 0, \tag{38}$$

(ii) RC satisfied, (iii) $\| \widehat{h} - h_0 \| \xrightarrow{p} 0$, *(iv)* $\frac{1}{n}\Sigma_{i=1}^n \| \widehat{\alpha}(W_i) - \alpha(W_i) \|^2 \xrightarrow{p} 0$ *and (v)* $\| g(W,\mu,h) - g(W,\mu_M,h_0) \| \leq b(W)(\| \mu - \mu_M \| + \| h - h_0 \|)$ *while* $E[b^2(W)] < \infty$, *then under the true DGP F_n: (i)* $S^2_{j,n} \xrightarrow{p} \sigma^2_j(\theta, F_n)$, *(ii)* $\| \widehat{\Sigma} - \Sigma(F_n) \| \xrightarrow{p} 0$ *and (iii)* $\| \widehat{\Omega} - \Omega(F_n) \| \xrightarrow{p} 0$.

Proof. Following the proof of Lemma 1, under assumption (ii), $\psi(W) = g(W,\mu_M,h_0) + \alpha(W)$, where $g(W,\mu_M,h_0) = m(W,\theta,h_0) - \mu_M$, $\alpha(W) = E\big(\frac{\partial m(W,\theta_0,h_0(x)}{\partial h_0(x)} \big| x\big)(y - h_0(x))$ and $E[\psi(W)] = 0$. Under assumption (ii), in particular RC assumption 3-7, and (iii), by Lemma 5.2 of Newey (1994a), $\widehat{\mu} \xrightarrow{p} \mu_M$. Then by assumption (iii) and (v), $\frac{1}{n}\Sigma_{i=1}^n \| g(W,\widehat{\mu},\widehat{h}) - g(W,\mu_M,h_0) \|^2 \xrightarrow{p} 0$, applying Lindeberg-Lévy CLT and continuous mapping theorem. Then by assumption (ii) and (iv), following the proof of Lemma 8.3 of Newey and McFadden (1994),

$$\frac{1}{n}\Sigma_{i=1}^n \widehat{\psi(W_i)}\widehat{\psi(W_i)}' \xrightarrow{p} \frac{1}{n}\Sigma_{i=1}^n \psi(W_i)\psi(W_i)',$$

where $\widehat{\psi_j(W_i)} = g_j(W_i,\widehat{\mu},\widehat{h}) + \widehat{\alpha}_j(W_i)$ and $\widehat{\alpha}_j(W_i) = E\big(\frac{\partial \widehat{m_j(W,\theta_0,h_0(x))}}{\partial h_0(x)} \big| x\big)(y_i - \widehat{h}_0(x_i))$. Thus,

$$\frac{1}{n}\Sigma_{i=1}^n \widehat{\psi_j(W_i)}\widehat{\psi_j(W_i)}' \xrightarrow{p} \frac{1}{n}\Sigma_{i=1}^n \psi_j(W_i)\psi_j(W_i)'$$

and

$$\frac{1}{n}\Sigma_{i=1}^{n}\widehat{\psi_l(W_i)}\widehat{\psi_q(W_i)}' \xrightarrow{p} \frac{1}{n}\Sigma_{i=1}^{n}\psi_l(W_i)\psi_q(W_i)',$$

where $l \neq q$. Then by assumption (i) and Lemma 3,

$$\frac{1}{n}\Sigma_{i=1}^{n}\psi_j(W_i)\psi_j(W_i)' \xrightarrow{p} E_{F_n}[\psi_j(W)\psi_j(W)'] = \sigma_j^2(\theta, F_n),$$

which leads to

$$S_{j,n}^2 = \frac{1}{n}\Sigma_{i=1}^{n}\widehat{\psi_j(W_i)}\widehat{\psi_j(W_i)}' \xrightarrow{p} \sigma_j^2(\theta, F_n)$$

by continuous mapping theorem. Regarding (ii), we need further the consistency of the off-diagonal element of $\widehat{\Sigma}$ under a dynamic DGP true F_n when we don't have the corresponding uniform integrability condition as (38). Then also by continuous mapping theorem, it's sufficient to show, under F_n, $\frac{1}{n}\Sigma_{i=1}^{n}\psi_l(W_i)\psi_q(W_i) \xrightarrow{p} E_{F_n}[\psi_l(W)\psi_q(W)']$. Let $Z_{n,i} \equiv \psi_l(W_{n,i})\psi_q(W_{n,i})$, following the proof of Lemma S.7.1 of Romano and Shaikh (2012b), by the inequality

$$|a||b|\mathbb{1}\{\{|a||b| > \lambda\} \leq |a|^2\mathbb{1}\{|a| > \sqrt{\lambda}\} + |b|^2\mathbb{1}\{|b| > \sqrt{\lambda}\},$$

we will have

$$\lim_{\lambda\to\infty} limsup_{n\to\infty} E_{F_n}\big[|Z_{n,i} - E_{F_n}(Z_{n,i})|\mathbb{1}\{|Z_{n,i} - E_{F_n}(Z_{n,i})| > \lambda\}\big] = 0.$$

Therefore, by Lemma 11.4.2 of Romano and Lehmann (2005), the desired result follows. (iii) is a direct consequence of (i) and (ii) by continuous mapping theorem so that $S_{j,n} \xrightarrow{p} \sigma_j(\theta, F_n)$. □

Lemma 5.

$$\lim_{\lambda\to\infty} sup_{F\in\mathbf{F}} E_F\big[\big(\frac{\psi_j(W)}{\sigma_j(F)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(F)}| > \lambda\}\big] = 0$$

if and only if $\forall F_n \in \mathbf{F}, \lim_{\lambda\to\infty} limsup_{n\to\infty} E_{F_n}\big[\big(\frac{\psi_j(W)}{\sigma_j(F_n)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(F_n)}| > \lambda\}\big] = 0.$

Proof. (only if): as $\lambda \to \infty$,

$$\begin{aligned} & sup_{F\in\mathbf{F}} E_F\big[\big(\tfrac{\psi_j(W)}{\sigma_j(F)}\big)^2\{\mathbb{1}\{|\tfrac{\psi_j(W)}{\sigma_j(F)}| > \lambda\}\big] \\ \geq\ & \lim_{n\to\infty} E_{F_n}\big[\big(\tfrac{\psi_j(W)}{\sigma_j(F_n)}\big)^2\{\mathbb{1}\{|\tfrac{\psi_j(W)}{\sigma_j(F_n)}| > \lambda\}\big] \\ \geq\ & 0 \end{aligned}$$

$\forall F_n \in \mathbf{F}$. The desired result followed by squeeze theorem when

$$\lim_{n\to\infty} E_{F_n}\big[\big(\frac{\psi_j(W)}{\sigma_j(F_n)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(F_n)}| > \lambda\}\big] = limsup E_{F_n}\big[\big(\frac{\psi_j(W)}{\sigma_j(F_n)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(F_n)}| > \lambda\}\big].$$

(if): Suppose the supremum exists for $E_F\big[\big(\frac{\psi_j(W)}{\sigma_j(F)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(F)}| > \lambda\}\big]$, then there exists a subsequence $\{F_n : F_n \in \mathbf{F}\}$ such that

$$\sup_{F\in\mathbf{F}} E_F\big[\big(\frac{\psi_j(W)}{\sigma_j(F)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(F)}| > \lambda\}\big] = \lim_{n\to\infty} E_{F_n}\big[\big(\frac{\psi_j(W)}{\sigma_j(F_n)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(F_n)}| > \lambda\}\big].$$

Thus $\underset{n\to\infty}{limsup} E_{F_n}\big[\big(\frac{\psi_j(W)}{\sigma_j(F_n)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(F_n)}| > \lambda\}\big] = \underset{F\in\mathbf{F}}{\sup} E_F\big[\big(\frac{\psi_j(W)}{\sigma_j(F)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(F)}| > \lambda\}\big] = 0$ as $\lambda \to \infty$. □

Lemma 6. *Let $J_n(x,F)$ be the distribution of the test statistic T_n^{min} as defined by (10) with $\widehat{\mu} = \frac{1}{n}\Sigma_{i=1}^n m(W_i,\theta,\widehat{h})$. Suppose all the assumptions in Lemma 4 satisfied. Let $\mathbf{F}'$ be the set of all distributions on $\mathbb{R}^p$. Define $\forall (Q,F) \in \mathbf{F}' \times \mathbf{F}$,*

$$\rho(Q,F) = max\Big\{\max_{1\le j\le k}\Big\{\int_0^\infty |r_j(\lambda,Q) - r_j(\lambda,F)|exp(-\lambda)d\lambda\Big\}, \parallel \Omega(Q) - \Omega(F) \parallel \Big\}$$

where $\forall 1 \le j \le k$,

$$r_j(\lambda,F) = E_F\Big[\Big(\frac{\psi_j(W)}{\sigma_j(F)}\Big)^2 \mathbb{1}\Big\{\Big|\frac{\psi_j(W)}{\sigma_j(F)}\Big| > \lambda\Big\}\Big].$$

Then for all sequences $\{Q_n \in \mathbf{F}' : n \ge 1\}$ and $\{F_n \in \mathbf{F} : n \ge 1\}$ that satisfy $\rho(Q_n,F_n) \to 0$,

$$\underset{n\to\infty}{limsup}\sup_{x\in\mathbb{R}} |J_n(x,Q_n) - J_n(x,F_n)| = 0. \tag{39}$$

Proof. For all sequences $\{Q_n \in \mathbf{F}' : n \ge 1\}$ and $\{F_n \in \mathbf{F} : n \ge 1\}$ that satisfy $\rho(Q_n,F_n) \to 0$, following the proof of Lemma S.12.1 of Romano and Shaikh (2012b), by assumption of the uniform integrability (38) and proof by contradiction, we would have

$$\lim_{\lambda\to\infty} \underset{n\to\infty}{limsup}\, r_j(\lambda,F_n) = 0$$

$$\lim_{\lambda\to\infty} \underset{n\to\infty}{limsup}\, r_j(\lambda,Q_n) = 0$$

by Lemma 5, which is the uniform integrability condition. Then suppose (39) fails, then there exists a subsequence n_v such that $\Omega(F_{n_v}) \to \Omega^*$, $\Omega(Q_{n_v}) \to \Omega^*$ and

either

$$\underset{x\in\mathbb{R}}{sup}|J_{n_v}(x,F_{n_v})-\Phi_{\Omega^*}(x,\ldots,x)|\nrightarrow 0$$

or

$$\underset{x\in\mathbb{R}}{sup}|J_{n_v}(x,Q_{n_v})-\Phi_{\Omega^*}(x,\ldots,x)|\nrightarrow 0$$

where Φ_Σ is the cdf of $\mathcal{N}(0,\Sigma)$. Consider under F_{n_v},

$$P\big(n^{-\frac{1}{2}}\Sigma_{i=1}^n\psi_1(W_i)\leq-\sigma_1(F_{n_v})x,\ldots,n^{-\frac{1}{2}}\Sigma_{i=1}^n\psi_k(W_i)\leq-\sigma_k(F_{n_v})x\big)$$

where $E_{F_{n_v}}[\psi_j(W)]=0\;\forall 1\leq j\leq k$. $(-\infty,-\sigma_1(F_{n_v})x]\times\cdots\times(-\infty,-\sigma_k(F_{n_v})x]$ is convex. Then by Lemma 3.1 of Romano and Shaikh (2008) and continuous mapping theorem,

$$|\zeta_{F_{n_v}}(\cdot)-\Phi_{\Omega(F_{n_v})}(\cdot)|\to 0,$$

where $\zeta_{F_{n_v}}(\cdot)\equiv P\big(\frac{n^{-\frac{1}{2}}\Sigma_{i=1}^n\psi_1(W_i)}{\sigma_1(F_{n_v})}\leq(\cdot),\ldots,\frac{n^{-\frac{1}{2}}\Sigma_{i=1}^n\psi_k(W_i)}{\sigma_k(F_{n_v})}\leq(\cdot)\big)$ and $(\cdot)$ represents any elements in the $(-\infty,-x]^k$. Since $|\Phi_{\Omega(F_{n_v})}(\cdot)-\Phi_{\Omega^*}(\cdot)|\to 0$ by continuity,

$$|\zeta_{F_{n_v}}(\cdot)-\Phi_{\Omega^*}(\cdot)|\to 0$$

by triangular inequality. By Lemma 4 and continuous mapping theorem, $S_{j,n}\xrightarrow{p}\sigma_j(\theta,F_n),\forall 1\leq j\leq k$. Then

$$\big|P\big(\frac{n^{-\frac{1}{2}}\Sigma_{i=1}^n\psi_1(W_i)}{S_{1,n}}\leq(\cdot),\ldots,\frac{n^{-\frac{1}{2}}\Sigma_{i=1}^n\psi_k(W_i)}{S_{k,n}}\leq(\cdot)\big)-\Phi_{\Omega^*}(\cdot)\big|\to 0$$

by Slutsky's Theorem. Following Lemma 1, $\sqrt{n}(\widehat{\mu}_{M_j}-\mu_{M_j})=n^{-\frac{1}{2}}\Sigma_{i=1}^n\psi_j(W_i)+op(1)$.

$$\big|P\big(\frac{\sqrt{n}(\widehat{\mu}_{M_1}-\mu_{M_1}(F_{n_v}))}{S_{1,n}}\leq(\cdot),\ldots,\frac{\sqrt{n}(\widehat{\mu}_{M_k}-\mu_{M_k}(F_{n_v}))}{S_{k,n}}\leq(\cdot)\big)-\Phi_{\Omega^*}(\cdot)\big|\to 0$$

by Slutsky's Theorem. Thus,

$$\begin{aligned}J_n(x,F_{n_v}) &= 1-P\big(\min_{1\leq j\leq k}\tfrac{\sqrt{n}(\mu_{M_j}(F_{n_v})-\widehat{\mu}_{M_j})}{S_{j,n}}\geq x\big)\\ &= 1-P\big(\tfrac{\sqrt{n}(\widehat{\mu}_{M_1}-\mu_{M_1}(F_{n_v}))}{S_{1,n}}\leq(\cdot),\ldots,\tfrac{\sqrt{n}(\widehat{\mu}_{M_k}-\mu_{M_k}(F_{n_v}))}{S_{k,n}}\leq(\cdot)\big).\end{aligned}$$

Therefore,

$$J_n(x, F_{n_v}) \to 1 - \Phi_{\Omega^*}(\cdot) = \Phi_{\Omega^*}(\circ),$$

where $(\circ)$ represents any elements in the $(-\infty, x]^k$. Similarly, $J_n(x, Q_{n_v}) \to \Phi_{\Omega^*}(\circ)$. Then by Polya's Theorem, both $\underset{x\in\mathbb{R}}{sup}|J_{n_v}(x, F_{n_v}) - \Phi_{\Omega^*}(x, \ldots, x)|$ and $\underset{x\in\mathbb{R}}{sup}|J_{n_v}(x, Q_{n_v}) - \Phi_{\Omega^*}(x, \ldots, x)|$ converge to 0, which is a contradiction. □

Lemma 7. *Suppose all the assumptions in Lemma 6 satisfied. Then $M_n(\beta)$ defined by (12)*

$$\underset{n\to\infty}{liminf}\underset{F\in\mathbf{F}}{inf} P\{\mu \in M_n(\beta)\} \geq 1 - \beta.$$

Proof. Consider any sequence $\{F_n \in \mathbf{F} : n \geq 1\}$ as the true DGP of $W_{n,i}$ and denote $\widehat{F_n}$ as the empirical distribution of $W_{n,i}$. Then trivially $P_{F_n}\{\widehat{F_n} \in \mathbf{F}'\} \to 1$. By Lemma S.12.2 of Romano and Shaikh (2012b), $\int_0^\infty |r_j(\lambda, \widehat{F_n}) - r_j(\lambda, F_n)|exp(-\lambda)d\lambda \overset{P_{F_n}}{\to} 0$, $\forall 1 \leq j \leq k$. And by Lemma 4, $\| \widehat{\Omega} - \Omega(F_n) \| \overset{p}{\to} 0$. Hence, by continuous mapping theorem, $\rho(\widehat{F_n}, F_n) \overset{P_{F_n}}{\to} 0$. Then by Lemma 6 and Theorem 2.4 of Romano and Shaikh (2012a),

$$\underset{n\to\infty}{liminf}\underset{F\in\mathbf{F}}{inf} P\{J_n^{-1}(\alpha_1, \widehat{F_n}) \leq T_n^{min} \leq J_n^{-1}(1 - \alpha_2, \widehat{F_n})\} \geq 1 - \alpha_1 - \alpha_2$$

holds for any $\alpha_1 \geq 0$ and $\alpha_2 \geq 0$ satisfying $0 \leq \alpha_1 + \alpha_2 \leq 1$. The desired result follows immediately when $\alpha_1 = \beta$ and $\alpha_2 = 0$. □

Lemma 8. *Suppose all the assumptions in Lemma 6 satisfied. If $\frac{\sqrt{n}\mu_j(F_n)}{\sigma_j(F_n)} \to \infty \, \forall 1 \leq j \leq k$, then $P_{F_n}(M_n(\beta) \subseteq \mathbb{R}_+^k) \to 1$.*

Proof. By definition of $M_n(\beta)$, $\mu \in M_n(\beta)$ is such that $\frac{\sqrt{n}(\mu_j(F_n) - \widehat{\mu_j})}{S_j} \geq K_n^{-1}(\beta, \widehat{F_n})$ $\forall 1 \leq j \leq k$, where $K_n^{-1}(\beta, \widehat{F_n})$ is defined as (13), that is,

$$\mu_j \geq \frac{K_n^{-1}(\beta, \widehat{F_n}) \times S_j}{\sqrt{n}} + \widehat{\mu_j} = \frac{\sigma_j(F_n)}{\sqrt{n}}\left[\frac{\sqrt{n}(\widehat{\mu_j} - \mu_j(F_n)}{\sigma_j(F_n)} + \frac{\sqrt{n}\mu_j(F_n)}{\sigma_j(F_n)} + \frac{K_n^{-1}(\beta, \widehat{F_n}) \times S_j}{\sigma_j(F_n)}\right]$$

$\forall 1 \leq j \leq k$. By Lemma 2, $\frac{\sqrt{n}(\widehat{\mu_j} - \mu_j(F_n))}{\sigma_j(F_n)} = Op_{Fn}(1)$. By Lemma 4 and continuous mapping theorem, $\frac{S_j}{\sigma_j(F_n)} \overset{p_{F_n}}{\to} 1$. By Lemma 6, $K_n(x, \widehat{F_n}) \to \Phi_{\Omega^*}(\circ)$, where $(\circ)$ represents any elements in the $(-\infty, x]^k$. Thus, $K_n^{-1}(\beta, \widehat{F_n}) = Op_{Fn}(1)$. Therefore, when $\frac{\sqrt{n}\mu_j(F_n)}{\sigma_j(F_n)} \to \infty \, \forall 1 \leq j \leq k$, the desired result follows. □

Lemma 9. *Let $J_n(x,F)$ be the distribution of the test statistic T_n given by (4), (6) and (7) with $\widehat{\mu} = \frac{1}{n}\Sigma_{i=1}^n m(W_i,\theta,\widehat{h})$. After replacing $\sqrt{n}\mu$ by $\sqrt{n}\lambda_n$ in T_n, denote $J_n(x,\lambda_n,F)$ as the distribution of it, i.e.*

$$J_n(x,\lambda,F) = P\Big(S\big(S_n^{-1}(\sqrt{n}(\widehat{\mu}-\mu(F))) + S_n^{-1}\sqrt{n}\lambda, \widehat{\Omega_n}\big) \le x\Big)$$

Suppose all the assumptions in Lemma 4 satisfied. Define $\rho(Q,F)$ as in Lemma 6. Suppose for some $\emptyset \neq I \subseteq \{1,\dots,k\}$,

$$\frac{\sqrt{n}\mu_j(F_n)}{\sigma_j(F_n)} \to \delta_j$$

for all $j \in I$ and some $\delta_j \ge 0$ and

$$\frac{\sqrt{n}\mu_j(F_n)}{\sigma_j(F_n)} \to \infty$$

for all $j \notin I$. Then

$$P_{F_n}\big(T_n > J_n^{-1}(1-\alpha+\beta, \mu(F_n), \widehat{F_n})\big) \to \alpha - \beta.$$

Proof. Consider $T_n = \underset{1\le j\le k}{max} -\frac{\sqrt{n}\widehat{\mu_j}}{S_j}$. Under F_n, $\frac{\sqrt{n}\mu_j(F_n)}{S_j} = \frac{\sigma_j(F_n)}{S_j}\frac{\sqrt{n}\mu_j(F_n)}{\sigma_j(F_n)}$. By Lemma 4 and continuous mapping theorem, $\frac{S_j}{\sigma_j(F_n)} \xrightarrow{p_{F_n}} 1$. Thus, by assumption,

$$\frac{\sqrt{n}\mu_j(F_n)}{S_j} \xrightarrow{p_{F_n}} \delta_j$$

for all $j \in I$ and some $\delta_j \ge 0$ and

$$\frac{\sqrt{n}\mu_j(F_n)}{S_j} \xrightarrow{p_{F_n}} \infty$$

for all $j \notin I$. Hence,

$$\underset{1\le j\le k}{max} -\frac{\sqrt{n}\widehat{\mu_j}}{S_j} = \underset{j\in I}{max}\{-\frac{\sqrt{n}(\widehat{\mu_j}-\mu_j(F_n))}{S_j} - \frac{\sqrt{n}\mu_j(F_n)}{S_j}\} + op_{F_n}(1). \tag{40}$$

Under Q_n, $\frac{\sqrt{n}\mu_j(Q_n)}{S_j} = \frac{\sigma_j(Q_n)}{S_j}\frac{\sigma_j(F_n)}{\sigma_j(Q_n)}\frac{\sqrt{n}\mu_j(F_n)}{\sigma_j(F_n)}$. For all sequences $\{Q_n \in \mathbf{F}' : n \ge 1\}$

and $\{F_n \in \mathbf{F} : n \geq 1\}$ that satisfy $\rho(Q_n, F_n) \to 0$, following the proof in Lemma 6 and by Lemma 5,

$$\lim_{\lambda\to\infty} \sup_{Q_n\in\mathbf{F}'} E_{Q_n}\big[\big(\frac{\psi_j(W)}{\sigma_j(Q_n)}\big)^2\{\mathbb{1}\{|\frac{\psi_j(W)}{\sigma_j(Q_n)}| > \lambda\}\big] = 0.$$

Thus, similarly, $\frac{S_j}{\sigma_j(Q_n)} \xrightarrow{p_{Q_n}} 1$. At the same time, proof by contradiction leads to $\| \Omega(Q_n) - \Omega(F_n) \| \to 0$. Therefore, $\frac{\sigma_j(Q_n)}{\sigma_j(F_n)} \to 1$. Thus,

$$\frac{\sqrt{n}\mu_j(Q_n)}{S_j} \xrightarrow{p_{Q_n}} \delta_j$$

for all $j \in I$ and some $\delta_j \geq 0$ and

$$\frac{\sqrt{n}\mu_j(Q_n)}{S_j} \xrightarrow{p_{Q_n}} \infty$$

for all $j \notin I$. Hence,

$$\max_{1\leq j\leq k} -\frac{\sqrt{n}\widehat{\mu}_j}{S_j} = \max_{j\in I}\{-\frac{\sqrt{n}(\widehat{\mu}_j - \mu_j(Q_n))}{S_j} - \frac{\sqrt{n}\mu_j(Q_n)}{S_j}\} + o_{p_{Q_n}}(1). \tag{41}$$

Following the arguments in the proof of Lemma 6 and further by Slutsky's Theorem, $J_n(x, F_n) \to \Phi_{\Omega^*}(\circ)$, where $(\circ)$ represents any elements in the$(-\infty, x + \delta_j]^{|I|}$ for each $j \in I$ for some $\delta_j \geq 0$. Similarly, $J_n(x, Q_n) \to \Phi_{\Omega^*}(\circ)$. Therefore, applying proof by contradiction as in Lemma 6, for all sequences $\{Q_n \in \mathbf{F}' : n \geq 1\}$ and $\{F_n \in \mathbf{F} : n \geq 1\}$, if $\rho(Q_n, F_n) \to 0$,

$$\lim_{n\to\infty}\sup_{x\in\mathbb{R}}|J_n(x, Q_n) - J_n(x, F_n)| = 0.$$

Besides trivially, $P_{F_n}\{\widehat{F_n} \in \mathbf{F}'\} \to 1$. And by the same arguments in the proof of Lemma 7, $\rho(\widehat{F_n}, F_n) \xrightarrow{P_{F_n}} 0$. Then by Theorem 2.4 of Romano and Shaikh (2012a), the desired result follows. The case of T_n given by (6) and (7) can be proved similarly. □

C Proof of Theorem 1

Under H_0 and the assumption $h_0 = E[Y|X]$, its sufficient to show $\limsup_{n\to\infty} sup_{F\in\mathbf{F}} E_F[\phi_n] \leq \alpha$. The proof then follows that of Theorem 2.1 in Romano et al. (2014). Suppose (9) fails, then there exists a subsequence n_t and $\eta > \alpha$ such that $E_{F_{n_t}}(\phi_{n_t}) \to \eta$. Observe that

$$\begin{aligned} E_{F_{n_t}}(\phi_{n_t}) &= 1 - P_{F_{n_t}}\Big(\{M_{n_t}(\beta) \subseteq \mathbb{R}^k_+\} \cup \{T_{n_t}(\theta) \leq C_{n_t}(\widehat{1-\alpha+\beta}, \theta)\}\Big) \\ &\leq 1 - P_{F_{n_t}}\Big(\{M_{n_t}(\beta) \subseteq \mathbb{R}^k_+\}\Big) \end{aligned}$$

and that

$$E_{F_{n_t}}(\phi_{n_t}) \leq 1 - P_{F_{n_t}}\Big(\{T_{n_t}(\theta) \leq C_{n_t}(\widehat{1-\alpha+\beta}, \theta)\}\Big).$$

By Lemma 8, when $\frac{\sqrt{n_t}\mu_j(F_{n_t})}{\sigma_j(F_{n_t})} \to \infty \; \forall 1 \leq j \leq k$, then $P_{F_{n_t}}(M_{n_t}(\beta) \subseteq \mathbb{R}^k_+) \to 1$. Hence, $E_{F_{n_t}}(\phi_{n_t}) \to 0$, which is a contradiction.

Then consider the case when $\frac{\sqrt{n_t}\mu_j(F_{n_t})}{\sigma_j(F_{n_t})} \to \delta_j$ for all $j \in I$ and some $\delta_j \geq 0$ and $\frac{\sqrt{n_t}\mu_j(F_{n_t})}{\sigma_j(F_{n_t})} \to \infty$ for all $j \notin I$. By observing that $C_n(\widehat{1-\alpha+\beta}, \theta)$ is the critical value of $J_n^{-1}(1-\alpha+\beta, \lambda, \widehat{F_n})$ at level $1-\alpha+\beta$ when $\lambda_{min} = \underset{\lambda \in M_n(\beta) \bigcap \mathbb{R}^k_+}{arg\,sup} J_n^{-1}(1-\alpha+\beta, \lambda, \widehat{F_n})$, which is weakly greater than $J_n^{-1}(1-\alpha+\beta, \mu, \widehat{F_n}) = J_n^{-1}(1-\alpha+\beta, \widehat{F_n})$. Hence,

$$E_{F_{n_t}}(\phi_{n_t}) \leq P_{F_{n_t}}\Big(\{T_{n_t} \geq J_{n_t}^{-1}(1-\alpha+\beta, \widehat{F_{n_t}})\}\Big) + \beta,$$

where $0 < \beta < \alpha$. By Lemma 9, $P_{F_{n_t}}\big(T_{n_t} > J_{n_t}^{-1}(1-\alpha+\beta, \widehat{F_{n_t}})\big) \to \alpha - \beta$. Then $E_{F_{n_t}}(\phi_{n_t}) \to \alpha$, which is a contradiction. Therefore, the desired result follows.

Q.E.D.

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